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Interface morphogenesis with a deformable secondary phase in solid-state lithium batteries

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Data S1 and S2

Materials and Methods

Materials preparation

Lithium/sodium (Li/Na) foil fabrication. Li (MSE Supplies) and Na (Sigma-Aldrich) foils were combined in molar ratios from 2.5 to 20 mol% and stacked in a Li|Na|Li sandwich structure to enable accumulative roll bonding. Each Li|Na|Li sandwich was folded and cold-rolled repeatedly with stainless-steel rollers until the phases were mixed. The final thickness of the foil was controlled with cold rolling.

Lithium-indium (Li-In) alloy foil counter electrodes. The accumulative roll bonding process was employed to prepare Li-In alloy foils to serve as counter electrodes for electrochemical testing. The In foils were prepared by cold-rolling In pellets (99.99%, Kurt J. Lesker) to form foils, and then stacking with Li foils to make In|Li|In sandwich structures. The Li:In atomic ratio was controlled to be 1:3 (Li:In) to form a mixture of LiIn and In phases in the Li-In electrode. The In|Li|In stack was rolled and folded repeatedly to form a uniform mixture.

Lithium-sulfur composite fabrication. Sulfur cathode composites were prepared by ball-milling sulfur powder (Thermo Scientific, 325 mesh, 99.5%), Li_6PS_5Cl (Ampcera, ~1 µm particle size), and multi-walled carbon nanotubes (MWCNTs, Graphene Supermarket) with a S:LPSC:CNT weight ratio of 2:3:1 in a Fritsch Pulverisette 7 planetary ball mill. 1.2 g of the mixture of S, LPSC, and MWCNTs was added to a ZrO_2 jar with four 10-mm diameter ZrO_2 balls and 12 g of 1-mm diameter ZrO_2 balls. The mixture was prepared and sealed in the jar within an Ar-filled glovebox for the milling process. Milling was performed at 500 rpm over 24 cycles, with milling for 10 min and resting for 5 min per cycle. After the milling process, the jar was opened in an Ar-filled glovebox to collect the composite powder.

Cell fabrication

90 mg of Li_6PS_5Cl (~10 µm particle size, Ampcera) SSE powder was inserted into a 1-cm diameter polyether ether ketone (PEEK) die, and it was pressed at 440 MPa uniaxial pressure for 5 min to form a dense pellet. The pressed pellets were typically 650-700 µm thick. Li foil (MSE Supplies) served as a counter electrode for the electrochemical Li stripping and EIS experiments. Li metal and the Li/Na foils were punched out and attached to Ti plungers. The foils on plungers were then inserted into the PEEK die to form a cell stack comprising Li/Na|LPSC|Li. For contact formation, the cell stack was pressed with 60 MPa uniaxial pressure for 5 min. The Li metal counter electrode loading was 8.9-10.2 mg cm^{-2} (167-190 µm thick). The mass of the Li/Na electrode was between 7.6 and 10.2 mg cm^{-2}, corresponding to a thickness of 110-170 µm. Li and Na symmetric cells tested in fig. S9 were fabricated by the same method. For Cu symmetric cells, 10 µm-thick Cu discs (diameter: 1 cm) were affixed to both sides of a pre-pressed LPSC pellet (60 MPa) and pressed with 440 MPa uniaxial pressure (fig. S9). The Na half-cells in figs. S10 and S13 were fabricated using the same methods as the Li/Na|LPSC|Li cell. The thickness of Na foil was 27 µm. For the Cu half-cell in fig. S13, a 10 µm-thick Cu disc was mounted onto the pre-pressed LPSC pellet

(pressed at 60 MPa). Subsequently, the Cu foil-mounted pellet was fully pressed with 440 MPa uniaxial pressure. A Li disc was then attached on the other side of the pellet to complete the Cu half-cell fabrication.

Li-In foils served as counter electrodes in the experiments involving the statistical analysis of Li stripping (Fig. 1E) and cycling tests (Fig. 4). Li-In and Li/Na electrode disks (1 cm diameter) were attached to either side of the SSE pellet in the PEEK die using Ti plungers. The Li/Na|LPSC|Li-In stacks were uniaxially pressed with the same procedure as described above. The thickness of Li-In disks was ~140 µm. The sulfur composite served as the cathode to examine cycling performance in full cells. 90 mg of LPSC was pre-pressed with ~95 MPa uniaxial pressure in the PEEK die. After that, an appropriate amount of the sulfur composite powder (mass loading and N:P ratio given in caption of relevant figures in Supplemental Materials) was added on one side of the pellet and uniaxially pressed with 440 MPa to densify the pellet and the sulfur composite cathode. The Li/Na or Li electrode (1 cm diameter) was added on the other side of the pellet using the Ti plunger and then pressed with 60 MPa of uniaxial pressure to form interfacial contact.

Electrochemical characterization

For cell testing, the pressed cell stacks were sandwiched between two steel plates, and the stack pressure for cell testing was precisely controlled by tightening four bolts and nuts with springs at each corner using a digital torque wrench. The stack pressure applied to the cell was calibrated with a force sensor prior to the tests. The applied current density was determined by the diameter of Ti plunger (1 cm diameter; 0.785 cm^2).

The galvanostatic Li stripping and cycling tests were conducted with Landt Instruments and Arbin battery testers. EIS measurements were carried out with a Bio-Logic SP-200 potentiostat with a frequency range of 2 MHz to 0.2 Hz. Impedance spectra were measured periodically after stripping certain amounts of capacity from the working electrode. All electrochemical tests were carried out at room temperature (25 °C) in an Ar-filled glove box.

Materials characterization

Cryogenic focused ion beam (cryo-FIB) scanning electron microscopy (SEM) imaging. Cross-section images of the electrodes were captured using a Thermo Fisher Helios 5CX FIB-SEM equipped with a Ga ion source and a Quorum cryogenic stage. Li/Na|LPSC|Li cells were constructed as described above and electrochemically cycled with desired conditions. To extract the cell stack from the PEEK die, 1 cm-diameter Cu foils were attached to the top and bottom sides of the cell stack prior to uniaxial pressing. The cell stack was extracted from the PEEK die in an Ar-filled glove box and transferred into the SEM chamber, with a few seconds of air exposure. The sample stage was cooled to -140 °C prior to the ion beam etching to reduce detrimental interactions with the ion beam. The first milling cuts were performed using an ion accelerating voltage of 30 kV and beam current of 65 nA. The final polishing was done at 30 kV and 2.8

nA. Images were captured using an Everhart–Thornley secondary electron detector with an accelerating voltage of 5 kV and current of 0.34 nA. Energy dispersive X-ray spectroscopy analysis was performed for elemental distribution mapping at 10 kV voltage (EDAX Octane Elite).

Nanoindentation. The composite films were mounted onto SEM stubs with crystal bond inside an argon glove box, and then the sample surface was cleaned by a blade. To avoid exposure to air, the samples were transferred from the glovebox into the SEM chamber with a Vacushut sample transfer device (Vacushut; Agar Scientific). In situ nanoindentation experiment was conducted with a nanomechanical instrument (InSEM; Nanomechanics, Inc.) inside an SEM system (Quanta 200 SEM; FEI). A standard Berkovich tip was used with target strain rate of 0.05 s^{-1}. The depth limit was set as 0.5 μm and the tip was held at maximum load for 1 s.

Operando X-ray tomography SSB fabrication. SSB cells for operando X-ray tomography were assembled inside an Ar-filled glovebox with custom-made tomography-scale housings, which feature a 2-mm diameter Ti rod within a PEEK die. 7 mg of LPSC powder (~1 μm, Ampcera) was pressed at 300 MPa uniaxial pressure within the PEEK die for 5 min to form the SSE pellet. 2-mm diameter Li and Li/Na foils were added to both sides of the SSE pellet and pressed at 50 MPa for 5 min to construct a half cell. 5 MPa of stack pressure was applied to the cell using hex-head screws on both ends, and O-rings were set between the screws and the cell housing case to seal the SSB from atmosphere during X-ray imaging.

Synchrotron X-ray computed tomography and digital image processing. Operando X-ray CT images were collected at beamline 2-BM of the Advanced Photon Source at Argonne National Laboratory. X-ray scanning was conducted every 15 min during electrochemical cycling of the SSB half cell, and 1500 image projections were captured at uniform intervals along a 180° rotation of the cell for each scan. A monochromatic beam with 25.5 keV energy was used for collecting the projections, with each projection taking 400 ms to collect. A scintillator converted the projected X-ray beam into visible light that was then focused with a 2× magnification lens to produce a voxel size of 1.4 μm. The projected beam was then captured with an Oryx 5.0 MP Mono 10GigE detector and reconstructed into 3D image datasets at 2-BM using the Gridrec method in the TomoPy toolbox for Python.

The reconstructed X-ray tomography images were first converted from 32-bit single precision floating point format to unsigned 16-bit integer format using MATLAB v2023a. This was performed on a normalized scale specified by the maximum and minimum grayscale intensity value from all scans to ensure that voxels maintained the same relative intensity after format conversion. An edge-preserving Gaussian blur (bilateral filter) was applied to cropped images in the region of interest with a standard deviation of 3 to remove insignificant variations in intensity. Image segmentation was first performed in Ilastik version 1.4.0 to isolate the Li/Na composite from the SSE in each 3D dataset. Further segmentation was performed in MATLAB using intensity-based thresholding slice-by-slice to separate the Li and Na portions of the

composite. 3D visualizations of the segmented datasets were generated with ORS Dragonfly version 2022.2.0.1409. The overall image processing procedure for the simulation of Li and Na coverage changes is displayed in fig. S16.

Modeling

All methods associated with modeling are included in Supplementary Text Sections 6 and 7.

Supplementary Text

Supplementary Text Section 1. Interfacial Li transport descriptors of the Li/Na and pure Li electrodes

The capability of Li to transport to the SSE interface was evaluated by assuming that the Li dissolution from the working electrode can deplete Li near the SSE interface, which would enable Li transport from the bulk to the interface (*7, 23*). As seen in fig. S6A, the time to reach Li depletion (t_0) was measured at different current densities (i) with zero applied stack pressure. t_0 was determined as the stripping time required to reach a 0.15 V cutoff. The voltage profiles at this 0.15 V cutoff typically show steep slopes, indicating Li depletion near the SSE interface (fig. S6A). The corresponding interfacial Li transport descriptor (D), describing the capability of Li to be transported towards the SSE interface, was estimated based on Sand's equation as follows (*23, 34, 50*).

$$\frac{1}{\sqrt{t_0}} = \frac{2}{\sqrt{D \cdot \pi} \cdot c_0 \cdot F} i \quad \text{(S1)}$$

In this equation, c_0 and F are the initial Li content in the composite electrode and Faraday's constant, respectively. The $t_0^{-1/2}$ versus i plots from the different electrodes are shown in fig. S6b. Slopes of the linear regression lines (dashed lines in fig. S6B), representing $\frac{2}{\sqrt{D \cdot \pi} \cdot c_0 \cdot F}$, are listed in Table S4. D was estimated from each slope value as follows: $D = 0.67 \times 10^{-11}$, 0.84×10^{-11}, 0.98×10^{-11}, 1.68×10^{-11}, and 2.47×10^{-11} cm^2 s^{-1} for pure Li, 2.5% Na, 5% Na, 10% Na, and 20% Na, respectively. Sand's equation assumes a simplified model of one-dimensional chemical diffusion (*15, 50*). This differs from the actual conditions at the SSE interface since the Li transport towards the SSE interface involves multiple phenomena, including Li self-diffusion, plastic deformation, and adatom diffusion, resulting from the morphological evolution of three-dimensional structures (*4, 14*). To avoid misconceptions from these complexities, here D is termed as 'interfacial transport descriptor'. Despite this difference, the estimated descriptors align with the unidirectional Li stripping tests performed at low stack pressures shown in Fig. 1, accurately representing the enhanced Li transport capability of the Li/Na composite electrodes within this pressure range. However,

at higher stack pressures, this trend diverges from the stripping capacity measurements, likely due to additional factors, such as the increased deformability of Li and Na phases in the electrode.

Supplementary Text Section 2. Electrochemical behavior of the composite electrode interface

The Nyquist plots of Li and Na symmetric cells shown in Fig. S9A feature a partial semicircle spanning from high to mid-frequency range (to ~1 kHz) and a low-frequency tail. The semicircle at the higher frequency represents the bulk response of the SSE, including bulk and grain boundary resistances (*39, 51*). This partial semicircle can also reflect resistances arising from the electrode-SSE interfacial features, including interphase formation, charge transfer, and the extent of interface contact, in a frequency range lower than the bulk response (*36, 52, 53*). The evolution of these interfacial characteristics, which encompasses both chemical and mechanical aspects, can influence their visibility in the spectra depending on the specific experimental conditions.

The low-frequency tail is generally associated with ion transport behavior in the interfacial region (*38, 39, 51*). For the case of the Li-SSE interface, a short and low-angle sloping tail was observed, indicating that the Li^+ ions are electrochemically transferable into Li metal with negligible diffusion resistance. This tail might be associated with Li diffusion near the SSE interface (*53*). On the other hand, the Na-SSE interface displayed a steeply increasing and much longer tail, which is characteristic of Li^+ ion blocking behavior (*39, 54*). The angle of this extended tail was similar to the impedance profile seen for Cu/Cu symmetric cells, indicating capacitive blocking behavior of Na at the SSE interface similarly to Cu. Unlike Na and Li, the Nyquist plots for the Cu symmetric cells had greater impedance across all frequency ranges, indicating that the Cu foil poorly contacts the SSE compared to the deformable alkali metals (*39*).

The Li-LPSC interface can undergo chemical reactions due to the high reactivity of Li metal (*55, 56*), which was reflected in the Nyquist plot measured after 10 h of rest time in Fig. S9A. We attribute the shift of the curves to the right after resting to increased interfacial impedance due to the interphase growth. The interfacial resistance of the Li symmetric cell increased by about 5 Ω cm^2. This limited increase could be attributed to the self-terminating nature of interphase formation at the Li-LPSC interface (*56*). Similarly, the Na symmetric cell exhibited an increase of approximately 5 Ω cm^2 during the same resting period, which was comparable to the Li symmetric cell. This suggests that the Na-LPSC interface has similar interphase reaction kinetics to that of the Li-LPSC interface. The minor increase in interfacial resistance observed in the Li symmetric cell further confirms that the interfacial resistance increase in Fig. 2A originates from morphological instability during stripping at the Li-SSE interface.

To investigate the evolution of the interface during cycling, cyclic voltammetry (CV) was performed using Na half-cells (Na|LPSC|Li). Two different voltage ranges were chosen (-0.05-0.5 V and -0.05-0.15 V). This choice was made since the reactions between the incorporated Na and the LPSC SSE are influenced by the voltage range (see fig. S4). The Na|LPSC|Li cells were tested under a stack pressure of 8.1 MPa.

As shown in fig. S10A, C, both cells exhibited linear cathodic and anodic currents around the Li electrode potential, indicating effective Li plating/stripping on the Na working electrode. The CV profile from a Li|LPSC|Li symmetric cell demonstrates a constant resistance at the Li|SSE interface (fig. S10E). The cell scanned over the broader voltage range showed an oxidation peak beyond between 0.3 and 0.4 V, with a decrease in current density upon repeated scans. EIS profiles before and after each CV scan (fig. S10B) provide additional insight into the nature of interphase formation. The Nyquist plot before the CV scans exhibited a small semicircle at high frequencies (>1 kHz, (1) in fig. S10B(i)) along with an inclined tail at low frequencies (<1 kHz, (3) in fig. S10B(i)). The first semicircle likely represents several sources of interfacial resistance, including interphase resistance, charge transfer resistance, and resistance from inhomogeneous interfacial contact (*53*). The inclined tail is due to the Li-ion blocking nature of the Na-SSE interface (*39*). With repeated CV scans across this broader voltage range, the first semicircle expanded, and a new inclined slope appeared with intermediate frequencies, as denoted by (2) in fig. S10B. This impedance evolution indicates the formation of a new interphase through a reaction of the Na with the SSE, and this interphase acts as an additional diffusive layer with an ion-blocking boundary (represented by the portions of the spectra denoted (2) and (3), respectively) (*57*). These trends were absent in the CV performed within the narrower voltage range of -0.05-0.15 V (fig. S10C, D), where the CV profiles overlap without anodic reactions, and the EIS plots remain consistent without semicircle growth upon repeated scans.

As shown in the cross-sectional cryo-FIB-SEM image after Li stripping from a 10% Li/Na electrode up to a 0.5 V cutoff (fig. S12), the Na phase remained after Li stripping and occupied almost the entire electrode layer, with minor Li residue present. The SSE layer displayed negligible observable structural or morphological changes at this scale, and only a few pores were observed at the Na-SSE interface. This result suggests that reaction of the LPSC with Na does not cause structural damage to the SSE over the micron length scale, but it may result in the formation of sodiated interphase products such as Na_2S, Na_3P, and NaCl, analogous to Li compounds generated in sulfide SSEs (*58*,*59*,*60*).

These results demonstrate that there is an interphase-forming reaction around 0.3-0.4 V vs. Li/Li^+, and controlling the electrode potential to remain less than these potentials can enable highly reversible Li cycling without excess interphase growth at the Li/Na binary composite electrode interface. Future work to understand the chemical nature of interphase evolution with Na in contact with various Li-transporting SSEs is warranted.

Supplementary Text Section 3. Additional grayscale intensity analysis of the XCT data

Figure S15A shows a cropped image of the interface between the Li/Na electrode and the SSE, from the larger image displayed in Fig. 3A. The image shows two distinct regions, which are the Li/Na electrode (10% Na) in the upper half and the pelletized LPSC in the lower half. The average grayscale intensities of these regions was examined within the boxed areas shown in fig. S15A. The histogram in fig. S15B displays the grayscale intensity distributions for these areas, with mean values of 97.1±58.6 for the Li/Na electrode and 125.5±43.7 for the LPSC. Figure S15C and d show 2D surface plots of grayscale intensity for the marked regions in Fig. S15A. The Li/Na electrode exhibits greater topographical variations between peaks and valleys (i.e., greater standard deviation) compared to the LPSC layer. The contrast in the image results from differences in X-ray attenuation coefficients, which depend on atomic number and electron density (*40, 61*). The average grayscale intensity of the Li/Na electrode is lower than that of the LPSC layer due to the lower atomic number of Li+Na compared to LPSC. The brighter, laterally elongated features within the composite layer correspond to the Na, as observed in the SEM cross-sectional images (fig. S3 and Fig. 2E-G).

These grayscale features facilitate segmentation analysis at the Li/Na-SSE interfacial boundary when analyzing line-scanned grayscale intensity profiles, as shown in Fig. 3B. Figure S15C shows identical image analysis to that shown in Fig. 3B, but for a different region of the interface (the region shown in fig. S15A). This was carried out to provide further insight into the morphological changes at the interface. All line scans were conducted at the same location in the images, marked by red boxes. The red line in each intensity profile represents the electrode-SSE interface boundary. In the initial state (fig. S15E(i)), the profile below the red line, corresponding to the LPSC layer, shows less intensity variation, with an average intensity value of 126.4±24.0, while the mean value of the profile above the red line, corresponding to the composite electrode, is 99.9±63.0, indicating greater variation compared to the LPSC layer. This consistency with the histograms in fig. S15B helps to identify the boundaries of the two regions in the profiles.

To track the behavior of Na within the electrode during electrochemical cycling, the Na phase closest to the SSE interface is marked with green lines, while other Na domains located further from the interface are marked with blue lines in the intensity profiles (fig. S15E). Initially, the first Na phase was distant from the SSE boundary and separated by an intensity valley, corresponding to a Li layer (fig. S15E(i)). As Li was stripped to 1 mAh cm^{-2} capacity (fig. S15E(ii-iv)), the Li layer between the SSE and the Na phase diminished and the Na domains approached the interface, causing the Na contrast to be

conflated with the SSE (green lines in fig. S15E). A shoulder on the SSE boundary (the second green line in fig. S15E(ii-iv)) was observed before the Na domain of interest (the green line in fig. S15E(i)) was reached to the SSE boundary in this location, possibly indicating that another Na domain was morphologically extended from an adjacent location. During Li plating to 1.1 mAh cm^{-2} (fig. S15E(iv-vi)), these Na domains receded from the SSE interface and merged with each other (see green lines in fig. S15E(v, vi)), with the Li layer observed growing adjacent to the interface. This behavior aligns with the cross-sectional imaging results shown in Fig. 2F, G. The Na domains located farther from the SSE interface (blue lines in fig. S15E) also advanced or receded depending on the cycling state.

Supplementary Text Section 4. Li cycling tests with sulfur composite cathodes

The cyclability of the Li/Na electrodes was evaluated in full cells with sulfur composite cathodes, as shown in figs. S19 and S20. The sulfur composite electrodes were prepared by mixing sulfur as the active material with LPSC and carbon nanotubes (experimental details are provided in the Methods section). A 5% Na anode||sulfur cell was assembled with a stack pressure of 2.5 MPa, and its behavior was tested at a current density of 0.5 mA cm^{-2} (~0.06 C, 1C=1675 mA g_s^{-1}). 15.3 mg cm^{-2} of sulfur composite (33.3% active mass) was loaded into the cells, corresponding to a theoretical areal capacity of ~8.5 mAh cm^{-2}. The 5% Na electrode had a mass loading of 6.0 mg cm^2, resulting in an N/P ratio of 2.3. Under the same conditions, the pure Li||sulfur cell experienced short circuiting after 4 cycles (fig. S19). In contrast, as shown in Fig. S19A, C, the 5% Na cell sustained stable cycling for 150 cycles, initially delivering a discharge capacity of 3.35 mAh cm^{-2} (658 mAh g_s^{-1}). During the first 20 cycles, the discharge capacity increased to ~3.9 mAh cm^{-2} (766 mAh g_s^{-1}). The cell exhibited stable voltage profiles throughout cycling (fig. S19A), retaining nearly 60% of its discharge capacity at the first cycle after 150 cycles. In comparison, the pure Li cell initially delivered a discharge capacity of 2.12 mAh cm^{-2} (416.5 mAh g_s^{-1}) and showed overall cycling capacities of less than 2 mAh cm^{-2}, significantly lower than that of the 5% Na cell (fig. S19B).

The cycling behavior of Li/Na anode||sulfur cells was also evaluated at the lower stack pressure of 0.8 MPa using a current density of 0.5 mA cm^{-2} (~0.12 C), as shown in fig. S20. The pure Li and 10% Na||sulfur cells were assembled with a sulfur composite loading of ~7.3 mg cm^{-2}, corresponding to a theoretical capacity of ~4.1 mAh cm^{-2}. The pure Li cell short-circuited during the first cycle, with an initial discharge capacity of 0.59 mAh cm^{-2} (244 mAh g_s^{-1}) (Fig. S20C), whereas the 10% Na cell sustained cycling for 69 cycles. The 10% Na cell exhibited an initial discharge capacity of 1.62 mAh cm^{-2} (670.0 mAh g_s^{-1}) and increasing discharge capacity over the first 4-5 cycles, followed by a decline, achieving 66% capacity retention after 69 cycles.

Supplementary Text Section 5. Nanoindentation on individual Na and Li phases

To determine the hardness of the individual Li and Na phases in the composite anode, in-situ nanoindentation was performed at site-specific individual Na and Li regions within the 20% Na composite material. Nanoindentation was performed using a custom-built in-situ nanomechanical instrument, SEMentor, with a Berkovich indenter tip to a depth of 500 nm. Figure S23A, B shows two representative load *vs.* displacement curves and a corresponding SEM image acquired during the nanoindentation experiment. The SEM image in fig. S23B shows an 85˚-tilted view of the indenter tip hovering 6 µm above the sample surface. Figure S23C, D contains top-down SEM images that show the residual indentation impressions outlined by dashed circles on the Na and Li phases, respectively. Hardness (*H*) is then calculated as

$$H = \frac{P}{A_c} \qquad \text{(S2)}$$

where P is the indentation load measured by the instrument and A_c is the contact area measured from the SEM image. To calculate the hardness, we measured the A_c directly from the impression images. The calculated hardness and corresponding parameters A_c and P are listed in Table S6. The result shows that the pure Na phase is 54% softer than the Li in the composite anode. The residual nanoindentation impressions indicate the presence of so-called pile-up, or the additional extruded material around the nanoindentation boundary. Such pile-ups have been associated with greater-than-measured contact depths in nanoindentation experiments; for example, Fincher *et al.* (*18*) showed that the measured nanoindentation depth in Li metal that exhibits pile-up can be underestimated by 11%. The underestimation of contact depth will result in an overestimated hardness.

Supplementary Text Section 6. Density functional theory (DFT) calculations of Li diffusion characteristics

We performed density functional theory (DFT) calculations to estimate the Li diffusivity at Li-Na interfaces. All DFT calculations were performed using the Vienna Ab initio Simulation Package (VASP) (*62-65*), with the projector augmented-wave (PAW) method (*66, 67*). The Li ($1s^2\ 2s^1$) and Na ($2p^6\ 3s^1$) electrons were treated as valence electrons in the pseudopotentials. The generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional *(68)* was used, and the orbitals were expanded using a plane wave basis with cutoff energy of 520 eV. We first performed structural

relaxations on unit cells of Li ($Im\bar{3}m$) and Na ($Im\bar{3}m$), allowing the cell shapes, cell volumes, and atom positions to relax, until the force on each atom was below 0.001 eV/Å. The Brillouin zones were sampled using (8×8×8) Monkhorst-Pack *k*-point grids. Interface structures were constructed using the package *MPInterfaces* (*69*), which implements the lattice matching algorithm proposed by Zur *et al*. (*70*). We chose the (100) orientation as a representative plane for Li and Na. The Li(100)-Na(100) interface was constructed using 12 layers of Li and Na with no vacuum region. The [010] directions of the Li and Na slab were aligned at 18.4° from each other, resulting in an overall lattice mismatch of 3.53%. Structural relaxation was performed on the interface, allowing the cell shape, cell volume, and atom positions to relax, until the force on each atom was below 0.01 eV/Å. The Brillouin zone was sampled using a (3×3×1) gamma centered *k*-point grid.

Since the interface model has no empty sites for Li diffusion, we removed one Li atom at the interface to introduce a Li vacancy site (fig. S24A). For each Li atom at the interface, a defect structure with one Li vacancy was created by removing that Li atom. Structural relaxations were performed on the defect structures, allowing only the atomic positions to relax, until the force on each atom was below 0.05 eV/Å. Climbing image nudged elastic band (NEB) (*71, 72*) calculations were carried out to determine the migration barriers of a Li vacancy at the interface. We chose two migration pathways (fig. S24B) originating from the site with the lowest vacancy formation energy. We used a spring constant of 5 eV/Å^2 and three intermediate images between each Li site. NEB calculations were performed, allowing only the atomic positions to relax, until the force on each atom was below 0.05 eV/Å.

The diffusivity, D, of the Li vacancy was calculated as follows (*73, 74*):

$$D = \frac{n_p v_o l^2 c}{2d} \exp\left(\frac{S_m}{k_B}\right) \exp\left(-\frac{E_m}{k_B T}\right) \quad \text{(S3)}$$

where n_p is the number of equivalent diffusion pathways, v_o is the attempt frequency, l is the jump length, c is the site fraction of vacancies, $d = 2$ is the dimensionality of the diffusion, k_B is the Boltzmann constant, S_m is the migration entropy, E_m is the migration barrier energy, and T is the temperature. We make several simplifying assumptions to estimate the diffusivity. We assume S_m to be negligible compared to E_m, and $n_p = 4$. We set l to be the unit cell lattice constant of Li (3.5 Å (*75*)), and v_o to be the Debye approximation of the attempt frequency (6.8 THz (*76, 77*)).

Figure S24C shows the energy landscape of the two migration pathways considered for the Li(100)-Na(100) interface. The migration barriers for paths 1 and 2 are 0.707 eV and 0.653 eV, respectively. Although the migration barriers are high for the straight-line migration pathways considered, zigzag-like pathways that avoid high-energy configurations may have lower migration barriers. For Li migration on the Li(100) surface, Gaissmaier *et al.*(*78*) computed the migration barrier to range between 0.02 eV and 0.47 eV. The higher migration barriers at the Li-Na interface compared to the Li surface can be understood

by the presence of surrounding atoms at the interface that need to be pushed apart during diffusion. The room temperature diffusivity at the Li-Na interface was estimated as a function of migration barrier energy and vacancy concentration, as shown in fig. S24D. The interface diffusivity is highly sensitive to the migration barrier, decreasing from ~10^{-6} cm^2/s to ~10^{-15} cm^2/s as the migration barrier increases from 0.2 eV to 0.7 eV. In comparison, the estimated Li diffusivity of the Li(100) surface (*75*) ranges between ~10^{-3} cm^2/s and ~10^{-12} cm^2/s. These results indicate that Li diffusivity is lower at the Li-Na interface than on the Li surface. Li diffusivity and migration barriers on the Li surface and at the Li-Na interface are listed in Table S7.

Supplementary Text Section 7. Electro-chemo-mechanical model for interface evolution

The electro-chemo-mechanical model describing the transport, stress distribution, and anode interface evolution is described in detail below.

Transport in SSE

The ionic transport in the SSE is governed by electric potential gradients. The charge conservation of Li ions in the SSE is given by:

$$\nabla.(\kappa_i \nabla\phi_e) = 0 \tag{S4}$$

where κ_i is the ionic conductivity of SE and ϕ_e is the electric potential in SSE.

Reaction Kinetics

The Butler-Volmer equation is used to obtain the reaction distribution at the Li-SSE interface:

$$i_{rxn} = i_0 \left\{ exp\left(\frac{F\eta_\phi}{2RT}\right) - exp\left(-\frac{F\eta_\phi}{2RT}\right) \right\} \tag{S5}$$

where i_{rxn} is the reaction current density, i_0 is the exchange current density, η_ϕ is the electric overpotential, F is the Faraday's constant, R is the universal gas constant, and T is the temperature. The electric overpotentials at the anode interface is defined as:

$$\eta_\phi = \phi_s - \phi_e - U \tag{S6}$$

where ϕ_s is the electric potential in the metal anode, ϕ_e is the electric potential in the SSE, and U is the equilibrium potential.

At the top electrolyte boundary, a uniform current is applied. No flux boundary condition is applied for the potential at the left and right boundaries. The presence and growth of voids significantly alters the current distribution at the interface; the reaction current density vanishes in regions where Li is not in contact with the SSE (gap, $g > 0$). The stripping of Li at the anode interface in regions of contact leads to change in interface morphology which is given by:

$$\frac{\partial r}{\partial t} = \frac{-\, i_{rxn}\, \Omega_{Li}}{zF} \quad \text{(S7)}$$

where r is the height of Li stripped due to reaction, t is the time, i_{rxn} is the reaction current density, Ω_{Li} is the molar volume of Li, and z is the number of valence electrons.

Mechanics

The von Mises plasticity model is used to calculate the stress distribution and deformation of Li and Na. First, the stress distribution is obtained for a linear elastic isotropic medium using the deformation from the previous state:

$$\sigma_{elas} = \frac{2\nu\mu}{1-2\nu} tr(\varepsilon_{elas}) + 2\mu\varepsilon_{elas} \quad \text{(S8)}$$

where $\boldsymbol{\sigma}_{elas}$ is the Cauchy stress tensor, $\boldsymbol{\varepsilon}_{elas}$ is the elastic strain tensor, μ is the shear modulus, and ν is the Poisson's ratio of the SSE or Na.

The effective stress inside the plastic material is calculated using the elastic stress:

$$\sigma_{eff} = \sqrt{\frac{3\, \boldsymbol{s} \,.\, \boldsymbol{s}}{2}} \quad \text{(S9)}$$

where $\boldsymbol{s}$ is the deviatoric elastic stress tensor. The yield condition is then evaluated. Plastic deformation occurs if the effective stress (σ_{eff}) in a region exceeds the corresponding yield strength (σ_y):

$$f_y = \sigma_{eff} - \sigma_y > 0 \quad \text{(S10)}$$

The yield strength, σ_y is given by the strain hardening relationship:

$$\sigma_y = \sigma_0 + H.\varepsilon_{pl} \quad \text{(S11)}$$

Where H is the hardening modulus and ε_{pl} is the plastic strain.

The plastic strain is accordingly modified only for the regions where plastic deformation occurs:

$$\varepsilon_{pl} = \frac{f_y}{3\mu + H} \quad \text{(S12)}$$

and the new stress state is updated as:

$$\sigma = \sigma_{elas} - \frac{3\mu\varepsilon_{pl}}{\sigma_{eff}} s \quad \text{(S13)}$$

For elastic regions, ε_{pl} is 0 and therefore the stress is correctly calculated as σ_{elas}.

A uniform stack pressure of 1 MPa is applied at the top boundary of the electrode material and no displacement condition is applied at the left and right boundaries. The creep deformation over time is modeled using the equation *(10, 79)*:

$$\dot{\varepsilon} = C_r \left(\frac{p}{\sigma_y}\right)^n exp\left(\frac{-Q_c}{RT}\right) \quad \text{(S14)}$$

where $\dot{\varepsilon}$ is the strain rate, C_r is a material constant, p is the stress in the material, σ_y is the yield strength, n is the stress exponent, and Q_c is the creep activation energy. The equation can be further described as:

$$\dot{\varepsilon} = A\left(\frac{p}{\sigma_y}\right)^n \quad \text{(S15)}$$

where A is the effective creep constant. The creep deformation, u_c can then be calculated as:

$$u_c = A\left(\frac{p}{\sigma_y}\right)^n . \; u_{mech}.t \quad \text{(S16)}$$

where u_{mech} is elastoplastic deformation of the metal and t is the time.

Over time, the contact regions will evolve and change the stress distribution. The contact condition at the Li-SSE interface is given as:

$$g.p = 0, \quad g \geq 0, p \geq 0 \quad \text{(S17)}$$

where g is the gap between the surfaces and p is the pressure at the interface.

For regions with no contact: $g > 0$ and $p = 0$.

For regions with contact: $g = 0$ and $p > 0$.

Surface Diffusion

The void growth rate due to Li diffusion along the surface or interface is given as *(80-83)*:

$$\frac{\partial h_{diff}}{\partial t} = \frac{D_{Li}\,\Omega_{Li}\,\delta_b}{wRT}\cdot\frac{\partial \mu}{\partial s} \quad \text{(S18)}$$

where $\partial h_{diff}/\partial t$ is the increase in void height with time due to diffusion, D_{Li} is the diffusivity of Li, Ω_{Li} is the molar volume of Li, w is the void/Na width over which diffusion occurs, R is the universal gas constant, T is the temperature and δ_b is the surface tension thickness. The change in chemical potential along the surface, $\partial \mu/\partial s$, is dictated by the surface energy and stress gradients at the interface *(80,82,84)*.

Interface Evolution

The overall change in interface morphology during stripping is governed by reaction kinetics, mechanical deformation and surface diffusion:

$$h_f = h_i + r_{st} - u_{mech} + h_{diff} - u_c - d_{Li-SE} \quad \text{(S19)}$$

where h_f is the new interface height and h_i is the initial height. The height is altered by Li stripping (r_{st}), elastoplastic deformation of Li (u_{mech}), growth due to diffusion (h_{diff}), creep deformation over time (u_c) and the relative motion between Li and SE (d_{Li-SE}).

At points where contact is present:

If $\Delta h = h_f - h_i > 0$, contact is lost and $g_{new} > 0$

If $\Delta h = h_f - h_i < 0$, contact is lost and $g_{new} = 0$

The values of parameters used in the modeling framework are provided in Table S8. The mechanical properties for Li and Na have been taken from recent studies on elastic, plastic and creep behavior of both metals (*46, 79, 85*).

Figures

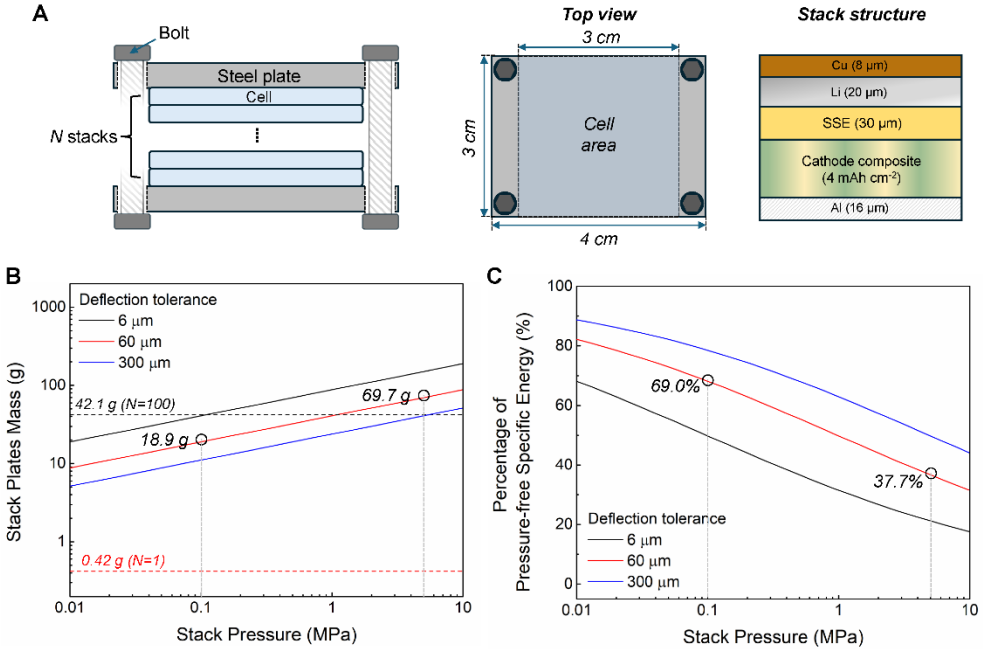


Fig. S1. The influence of stack pressure on the specific energy of cells with stack plates as a function of stack pressure. **(A)** Schematic illustration of an SSB cell assembled with *N* stacked layers sandwiched between two stack plates. A top-view image of the cell and the detailed cell structure are displayed on the right. **(B)** Stack plate mass as a function of applied stack pressure under different required deflection tolerance of the plates. For a smaller deflection tolerance, the stack plates must be thicker. The red- and black-dashed horizontal lines indicate the stack mass when *N*=1 and *N*=100, respectively. The required mass of the stack plates increases with increased stack pressure at a required deflection tolerance. **(C)** The percentage of pressure-free specific energy of the cell (when *N*=100) as a function of stack pressure for different deflection tolerances. This percentage is defined as the mass of the stack (*N* layers) divided by the combined mass of the stack and the stack plates; this percentage value represents the reduction of specific energy of the cell with the stack plates compared to just the stack. For simplicity, the mass of the bolts and cell casing is not included in the calculation. Assumptions for the calculation are provided in Table S1. For a single-layer cell (*N*=1), the plate mass far exceeds the cell mass (0.42 g for *N*=1 in (B)) regardless of stack pressure, yielding an extremely low specific energy of the cell. As another example, the mass of a multi-layer stack with *N*=100 is 42.1 g (B), and a 5 MPa pressure with 60 μm tolerance would require a plate mass of 69.7 g. The cell with stack plates thus exhibits only 37.7% of the pressure-free (or stack-level) specific energy (C). In contrast, reducing the stack pressure to a lower value, such as 0.1 MPa, allows for a decrease of the plate mass to 18.9 g (at 60 μm tolerance), making the housing lighter than the stack mass of 42.1 g (*N*=100). The specific energy of the cell is now 69.0% of that of the stack (C).

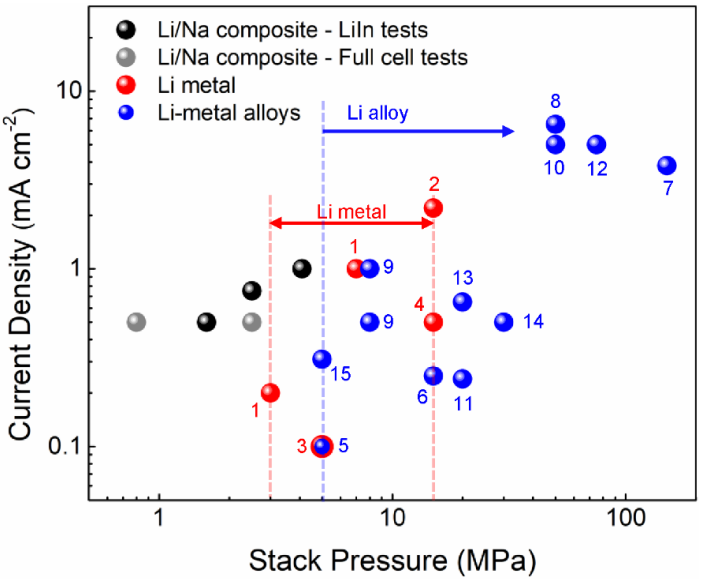


Fig. S2. Plot of current density and stack pressure comparing solid-state cells from this work (gray and black points) with other studies (red and blue points). All data in this plot were obtained using sulfide SSEs at or near room temperature. Current density and stack pressure values, as well as the literature references, are listed in Table S2.

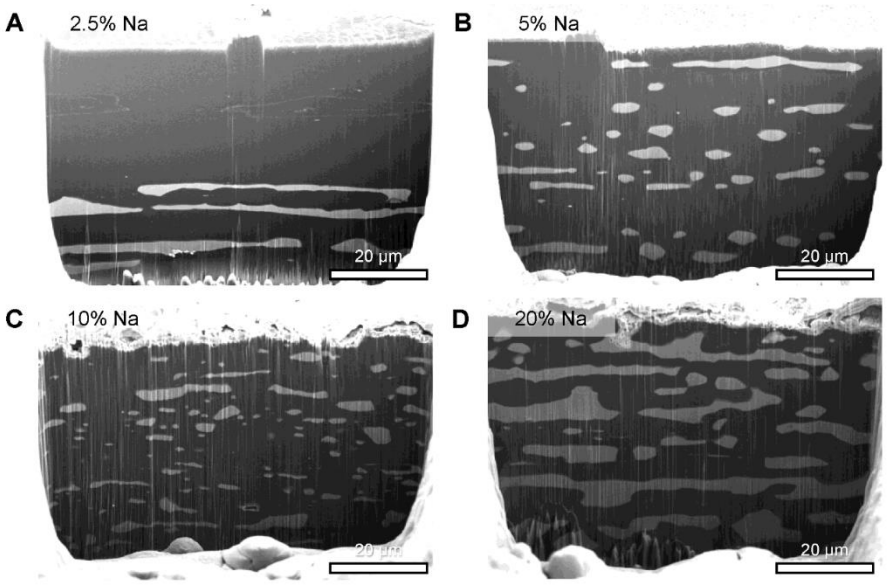


Fig. S3. Cryo-FIB-SEM cross-sectional images of the Li/Na foils with different Na contents: (**A**) 2.5% Na, (**B**) 5% Na, (**C**) 10% Na, and (**D**) 20% Na. The lighter domains are Na.

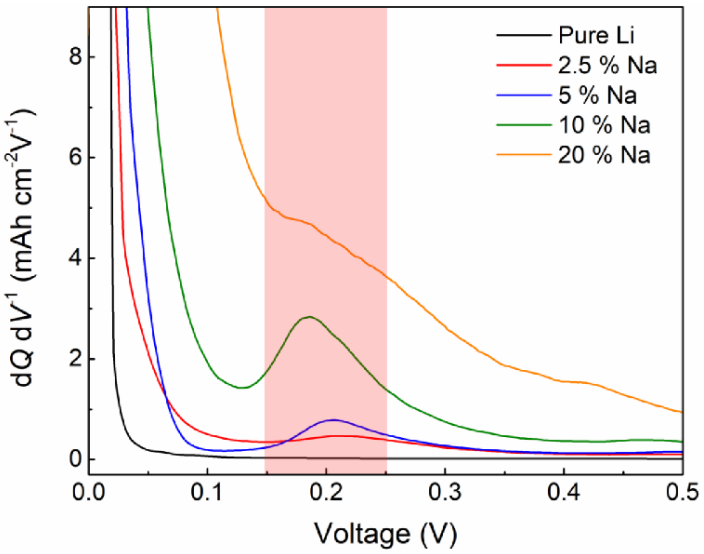


Fig. S4. Differential capacity (dQ dV^{-1}) profiles of the Li/Na electrodes and a pure Li electrode from the stripping voltage profiles at 0 MPa stack pressure shown in Fig. 1C. The hump region near 0.2 V indicates an interfacial side reaction of the Na.

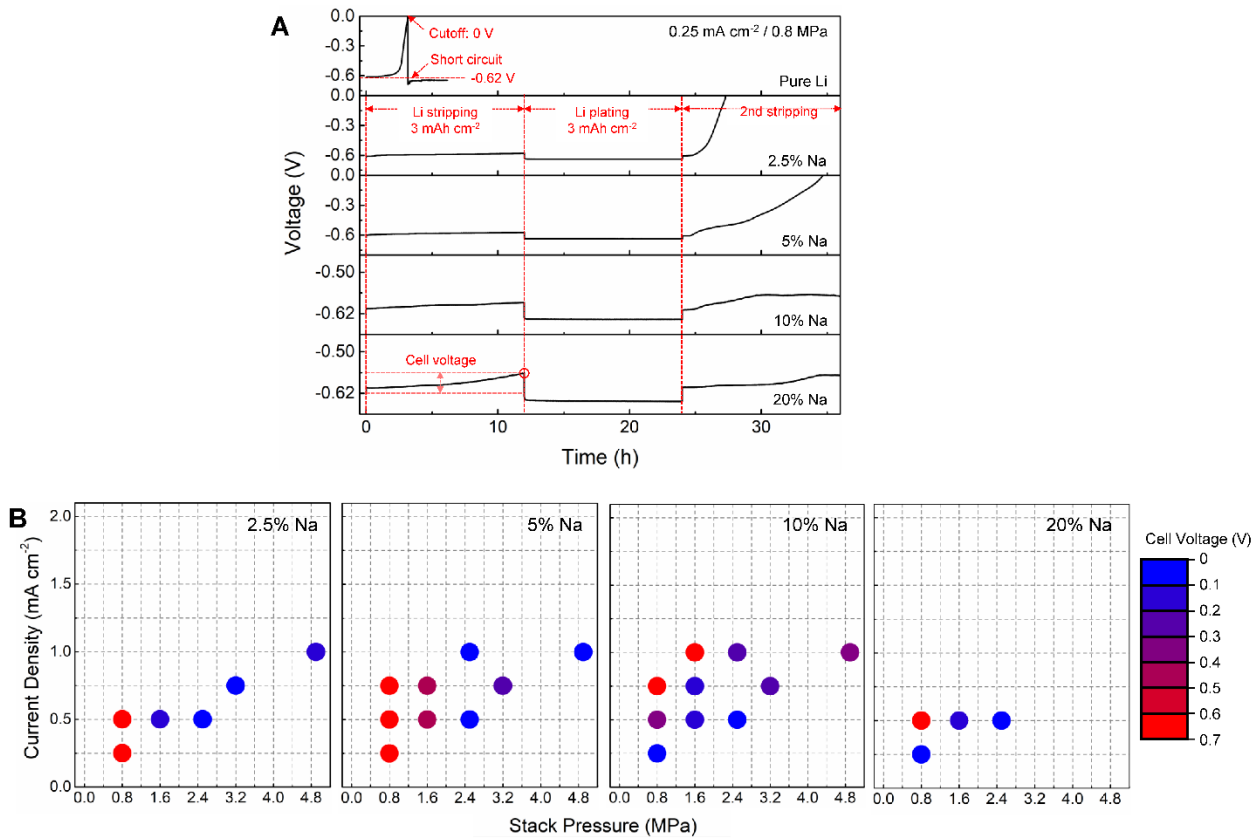


Fig. S5. Li stripping capability of Li/Na composite electrodes. (**A**) Galvanostatic cycling profiles of pure Li and Li/Na electrodes with 2.5% Na, 5% Na, 10% Na, and 20% Na (from top to bottom panels) at a current density of 0.25 mA cm^{-2} and a stack pressure of 0.8 MPa. (**B**) Color maps of ending voltage after the second Li stripping for 2.5-20% Na electrodes under varying stack pressures (0.8-4.8 MPa, *x*-axis) and current densities (0.25-2.0 mA cm^{-2}, *y*-axis). Li_1In_3 alloy foils (In+LiIn phases, +0.62 V vs. Li/Li^+) served as the counter electrodes in these tests, which were used to suppress short circuiting at the counter electrode at higher current densities (*86*, *87*). In (A), the working electrodes were initially stripped until either 3 mAh cm^{-2} was achieved or the cutoff voltage of 0 V was reached, followed by plating of 3 mAh cm^{-2}. If no short circuiting occurred during the first cycle, Li was then stripped again from the working electrode. The end stripping voltages, defined as the voltage difference between the open circuit voltage (-0.62 V) and the ending voltage, were plotted on the stack pressure-current density maps found in (B). Cell voltages from the second Li stripping step are presented in (B). Data are included in these maps only if the cell survived the first cycle without short circuiting. All pure Li electrode cells short-circuited during the plating step; hence, they are not shown in (B).

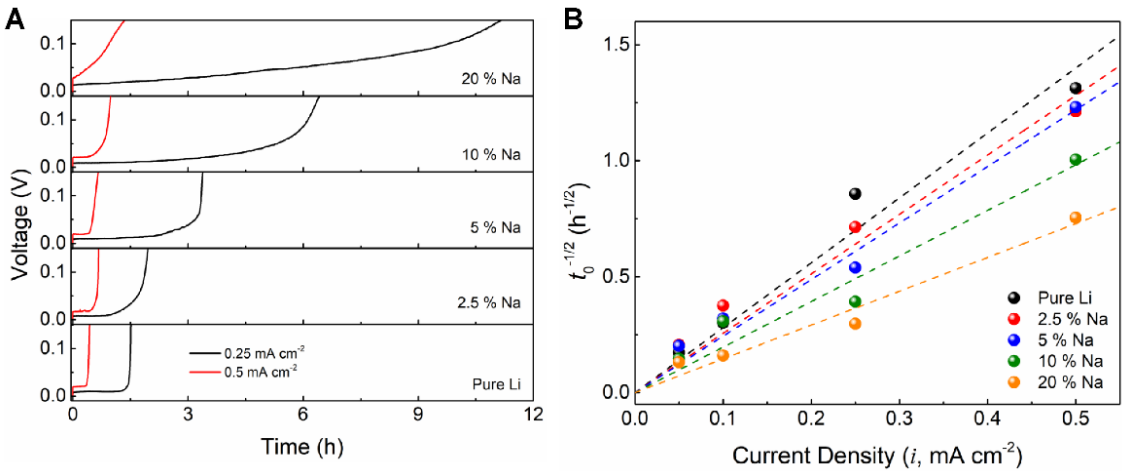


Fig. S6. (**A**) Time-dependent stripping voltage profiles for Li/Na electrodes and pure Li electrodes at current densities of 0.25 and 0.5 mA cm^{-2}. No stack pressure was applied to the cells, and only the pressure from the weight of cell components (e.g., plunger and stack plate), approximately 5 kPa, was exerted on the cell stack during the measurement. In these tests, the depletion time (t_0) was determined as the time when the voltage reached the 0.15 V cutoff. (**B**) $t_0^{-1/2}$ as a function of current density from the data in (A). The dashed lines indicate linear correlations between i and $t_0^{-1/2}$ based on Sand's equation, and their fitting parameters and the extracted Li transport descriptors are listed in Table S4. The current densities in this test were normalized by the geometric area of the electrode attached to the Ti plunger (0.785 cm^2).

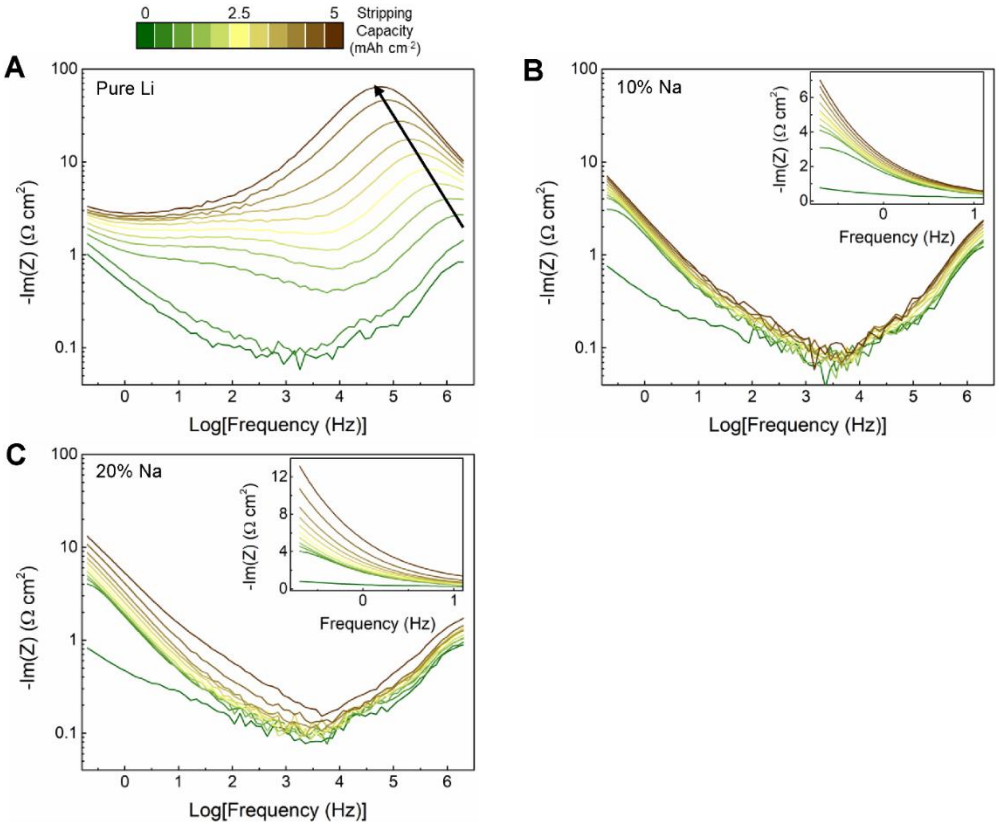


Fig. S7. Imaginary impedance *vs.* frequency plots of (**A**) pure Li, (**B**) 10% Na, and (**C**) 20% Na electrodes from in situ EIS analysis shown in Fig. 2A, B and Fig. S8C. The black arrow in (A) denotes a shift of the apex frequency in the high frequency range during the semicircle evolution shown in Fig. 2A. The observed apex frequency shift reflects changes in the impedance response due to interfacial morphology evolution (*26*, *36*, *37*). This evolution of interfacial morphology can lead to changes in the time constant associated with current constriction resistance, causing the apex frequency of this impedance feature to shift to lower values as the void size increases or interfacial contact is reduced. As compared to this, the 10% Na electrode in (B) showed similar profiles in the same frequency range during Li stripping. The inset graphs in (B, C) display the development of low-frequency tails over a linear scale of imaginary impedance.

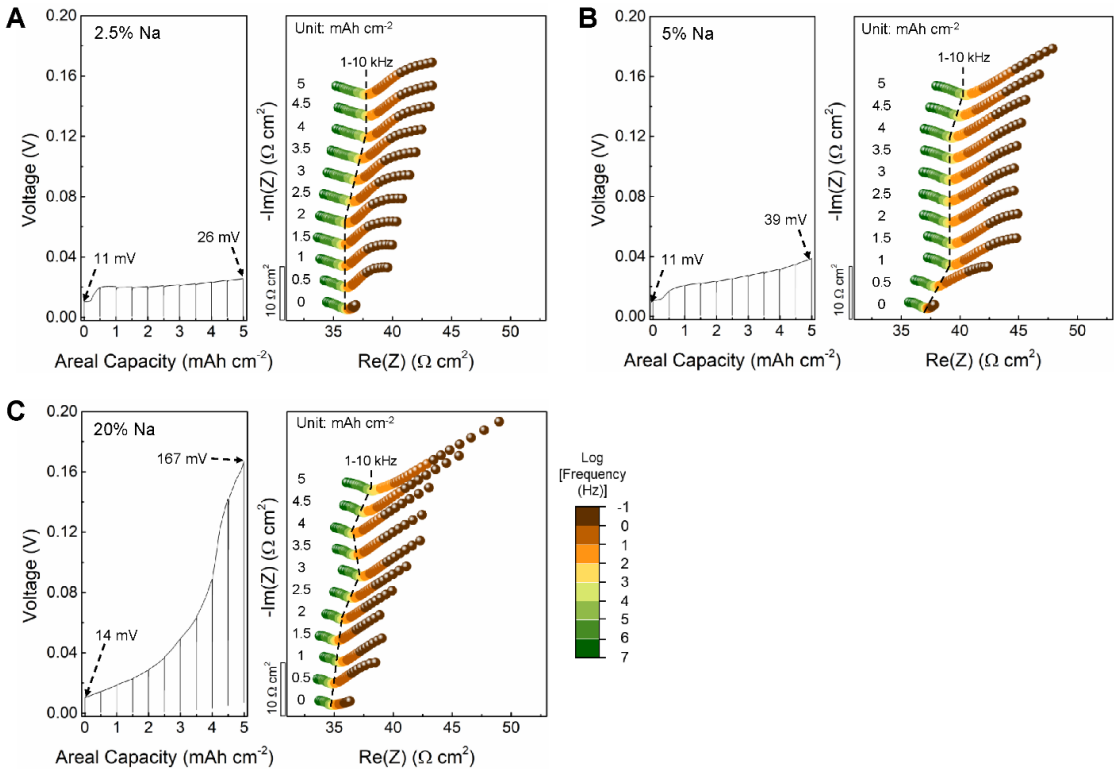


Fig. S8. In-situ EIS analysis of the (**A**) 2.5% Na, (**B**) 5% Na, and (**C**) 20% Na electrodes during electrochemical Li stripping. The left and right panels show the stripping voltage profiles and corresponding evolution of the Nyquist plots, respectively. All cells were tested at 1.6 MPa stack pressure in half cells, with Li metal as the counter electrode. Potentiostatic EIS spectra were collected every 0.5 mAh cm^{-2} during Li stripping with a current density of 0.25 mA cm^{-2}.

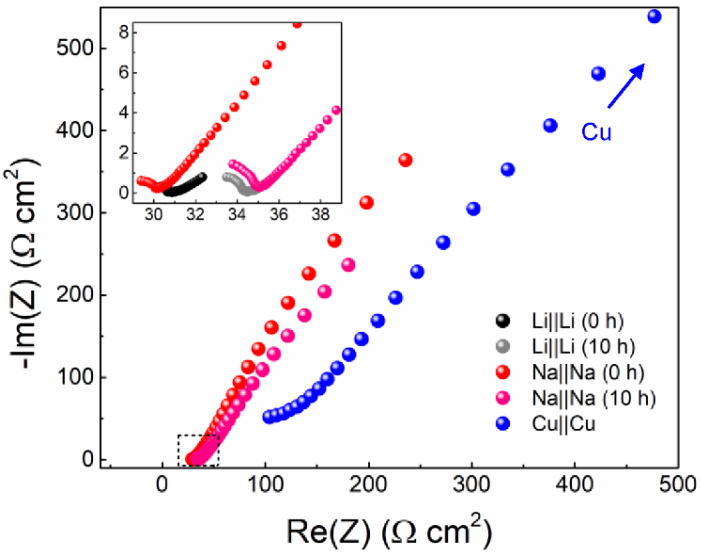


Fig. S9. Nyquist plots of Li, Na, and Cu symmetric cells in the initial state and after resting for 10 h. The dashed square area is enlarged in the inset to show the Nyquist plots of Li and Na symmetric cells. The EIS experiments for these symmetric cells were performed at a stack pressure of 1.6 MPa, which is identical to the stack pressure of the 10% Na cell in Fig. 2B. Details of the impedance behaviors were discussed in Supplementary Text 2.

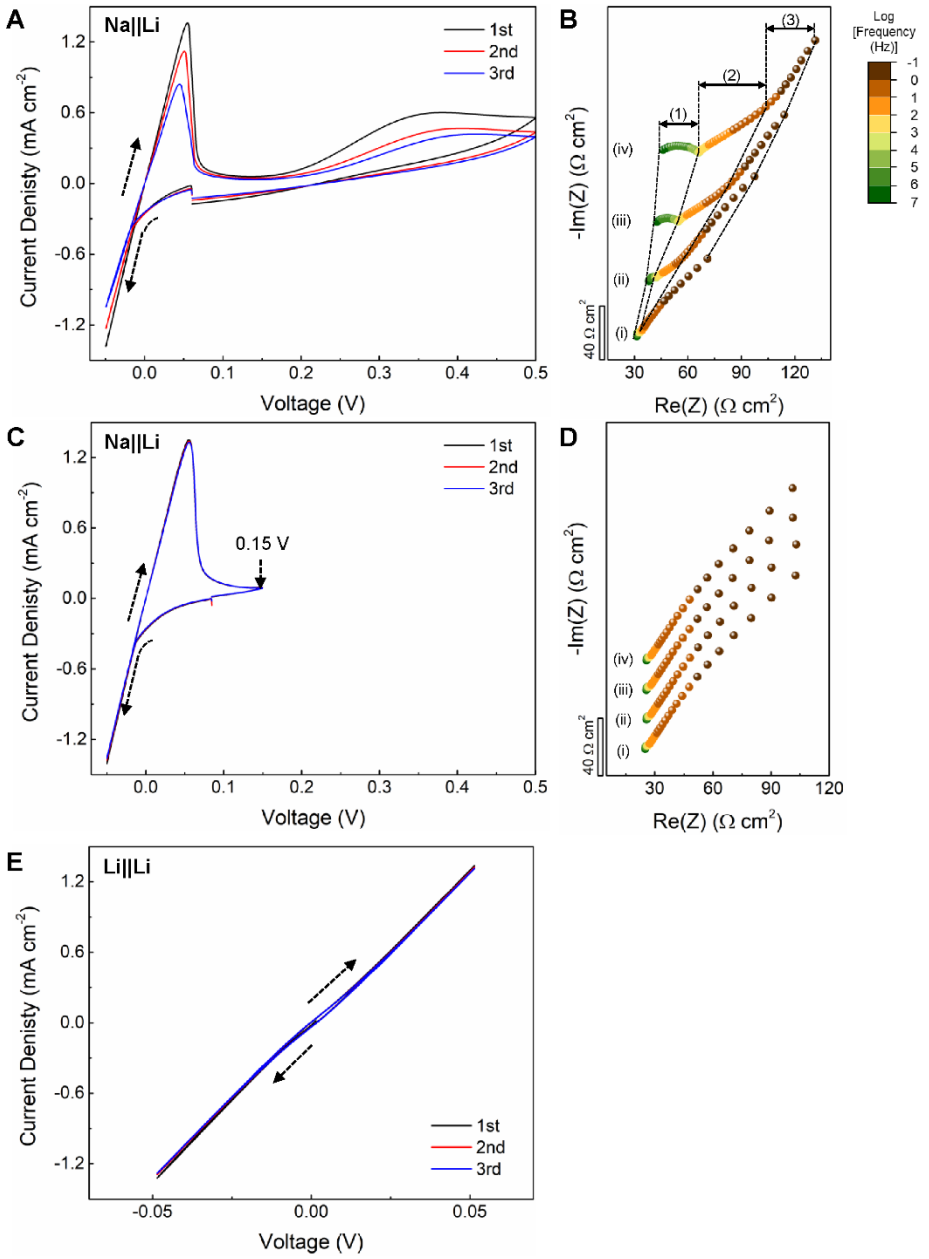


Fig. S10. (A) Cyclic voltammetry (CV) curves of a Na|LiPSCl|Li half-cell over a voltage range of -0.05 to 0.5 V. **(B)** Nyquist plots in the initial state and after each CV test for the cell in (A); (i) initial state, and after the (ii) 1st, (iii) 2nd, and (iv) 3rd cycle. **(C)** CV curves of a Na|LiPSCl|Li half-cell over a voltage range of -0.05 to 0.15 V. **(D)** Nyquist plots in the initial state and after each CV test for the cell in (C); (i) initial state, and after the (ii) 1st, (iii) 2nd, and (iv) 3rd cycle. The Na||Li half cells were tested under a stack pressure of 8.1 MPa. CV was performed at a rate of 1 mV s^{-1}, with the cathodic scan first. (**E**) CV curves of a symmetric Li||Li cell over a voltage range of -0.05 to 0.05 V.

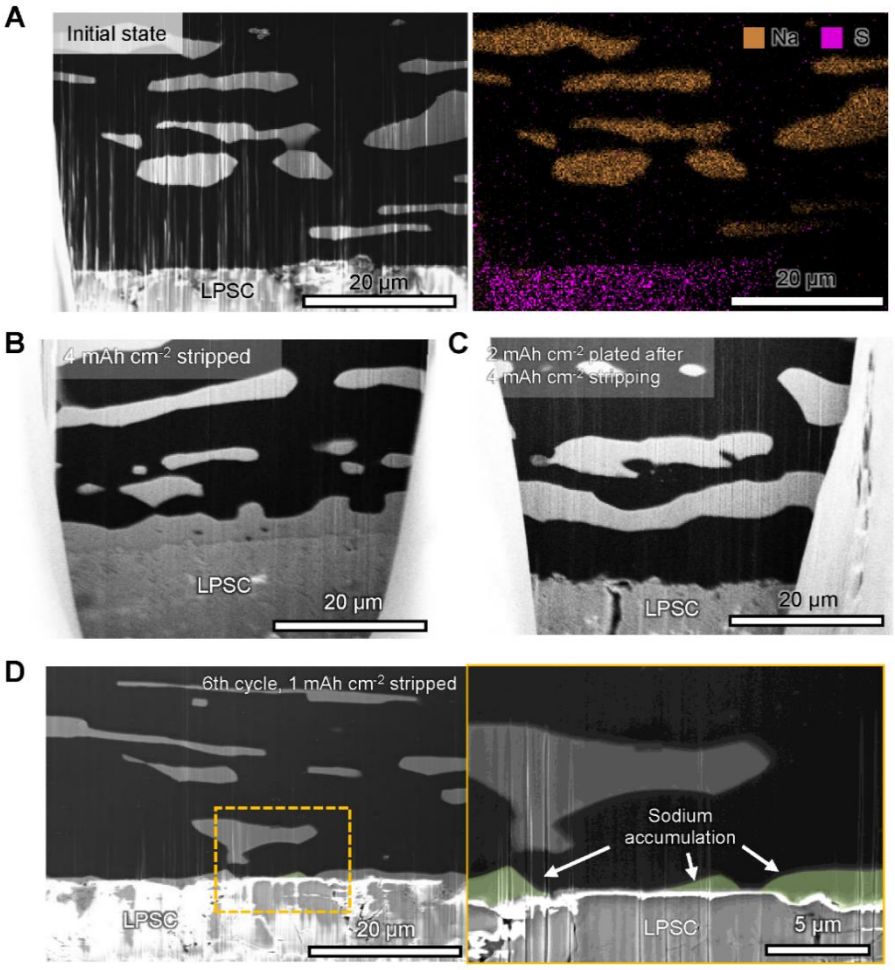


Fig. S11. Ex-situ cryo-FIB-SEM images of a 10% Na electrode-SSE interface in (**A**) the initial state before galvanostatic cycling, (**B**) after 4 mAh cm^{-2} stripping, (**C**) after 2 mAh cm^{-2} of Li deposition following 4 mAh cm^{-2} of stripping, and (**D**) after six stripping/plating cycles with 2 mAh cm^{-2} capacity followed by stripping to 1 mAh cm^{-2}. The right image in (A) shows an EDS elemental distribution map of Na and S. The pristine images in Fig. 2F, G are displayed in (B) and (C) without green-colored overlays. In (D), an enlarged image of the yellow box is shown on the right, and an artificial green color has been overlaid on Na domains that contact the SSE. All cells in (A-D) were tested with a stack pressure of 1.6 MPa and current density of 0.25 mA cm^{-2}.

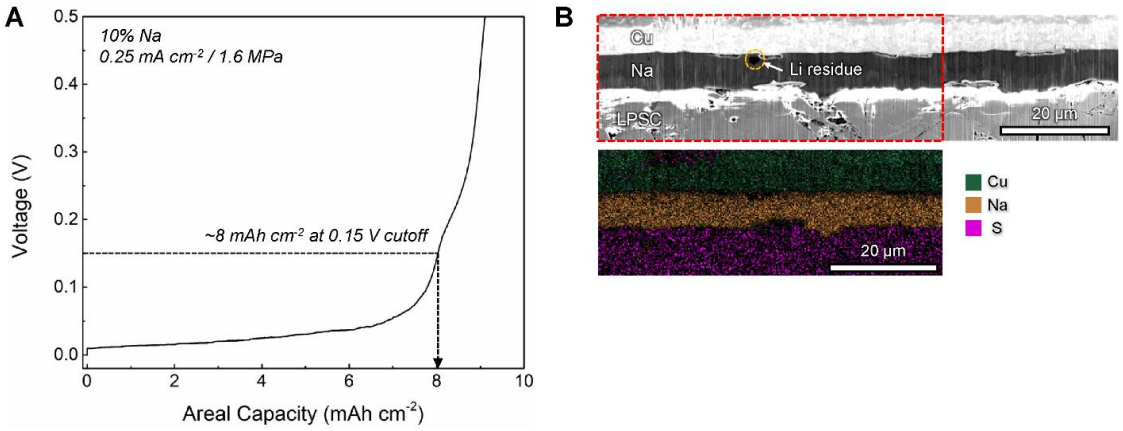


Fig. S12. (A) Voltage profile during Li stripping from an electrode with 10% Na using 0.25 mA cm^{-2} current density and 1.6 MPa stack pressure in a Li/Na|LPSC|Li cell configuration. The initial thickness of the electrode was 62 µm, corresponding to a theoretical Li capacity of 10.8 mAh cm^{-2}. When Li was stripped to a 0.15 V cutoff, approximately 8 mAh cm^{-2} of Li was removed, achieving ~74% utilization. (**B**) Ex-situ cryo-FIB-SEM image of the 10% Na electrode after the Li stripping process shown in (A). Li was intentionally stripped to a 0.5 V cutoff to observe the interfacial state. The EDS elemental distribution map of the region within the dotted red box shows the accumulated Na layer after Li was stripped from the electrode. The image in (B) reveals that the Na phase primarily remains in the electrode layer, with only minor residual Li. This result confirms that a significant amount of Li can be utilized without being trapped or hindered at the interface.

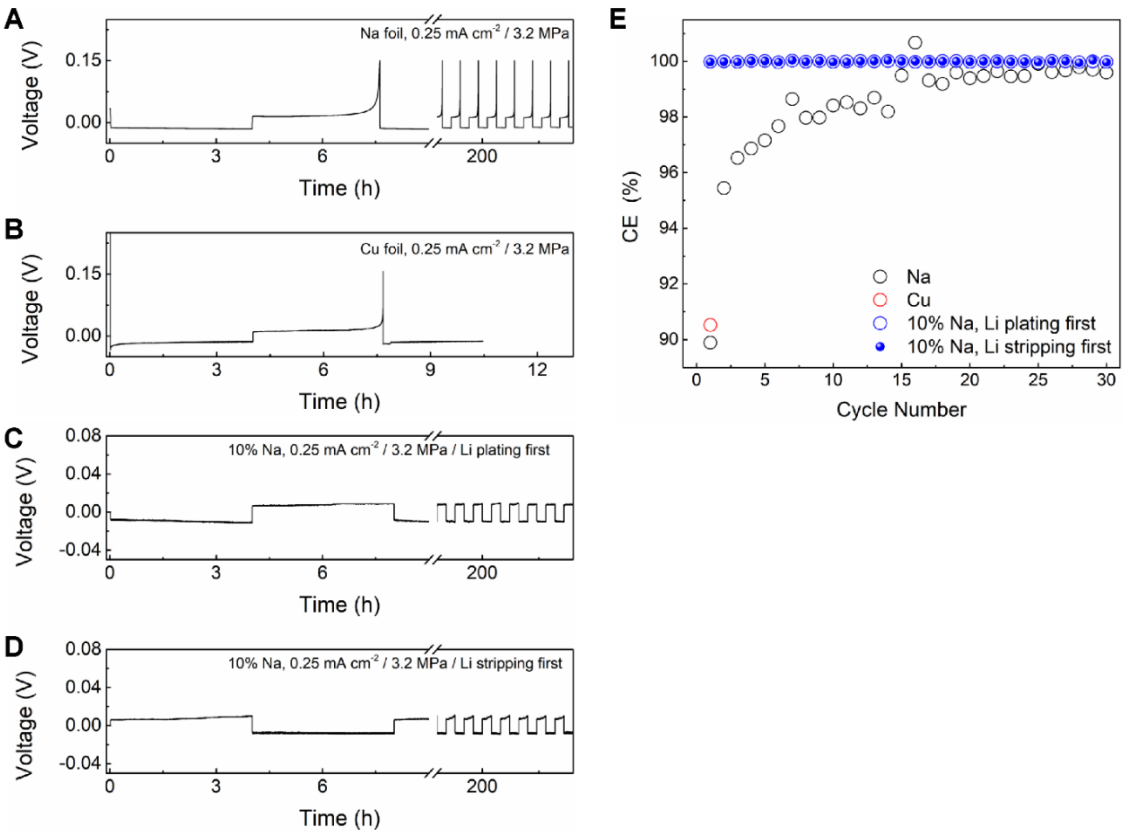


Fig. S13. Half cell tests for investigating Na metal as a current collector. **(A-D)** Voltage profiles of half cell tests using **(A)** a Na current collector (Na|SSE|Li), **(B)** a Cu current collector (Cu|SSE|Li), and **(C, D)** Li/Na composite working electrodes (Li/Na|SSE|Li). All cells were assembled with 3.2 MPa stack pressure and cycled at a current density of 0.25 mA cm^{-2} and a cycling capacity of 1 mAh cm^{-2}. The Na and Cu half cells were cycled with a 0.15 V cutoff to avoid Na reaction with the SSE (fig. S10). The Li/Na half cell in (C) began with Li plating on the working electrode first, while the Li/Na half cell in (D) began with Li stripping from the working electrode first. (E) Corresponding CEs during cell cycling. The initial CEs of the Na and Cu half cells were 89.9% and 90.5%, respectively. The Cu half cell short-circuited after 1 cycle, while the Na and Li/Na half cells demonstrated stable cycling over 30 cycles. The Na|SSE|Li cell exhibited an average CE of 98.5±2.0% over these 30 cycles. The half cells using the Li/Na electrode demonstrated higher CEs than the Na half cell with negligible capacity losses due to the excess Li present (average CEs of >99.9% were achieved for both).

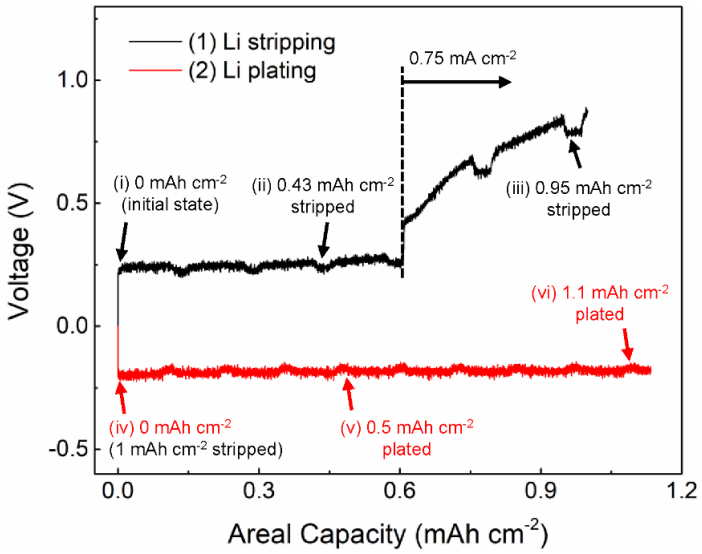


Fig. S14. Voltage profile obtained during the operando X-ray computed tomography experiment presented in the main text, using the miniature tomography cells. Li stripping from the composite electrode was first performed with a capacity of 1 mAh cm^{-2}, followed by plating into the composite electrode with a capacity of 1.13 mAh cm^{-2}. The cell was cycled at a current density of 0.5 mA cm^{-2}, with 0.75 mA cm^{-2} current density applied to cell after 0.6 mAh cm^{-2} of Li stripping. A 5 MPa stack pressure was applied to the cell. Electrochemical cycling states labeled in Fig. 3B and fig. S15E are marked on the graph. The periodic divots in the voltage correspond to X-ray exposure during tomographic imaging.

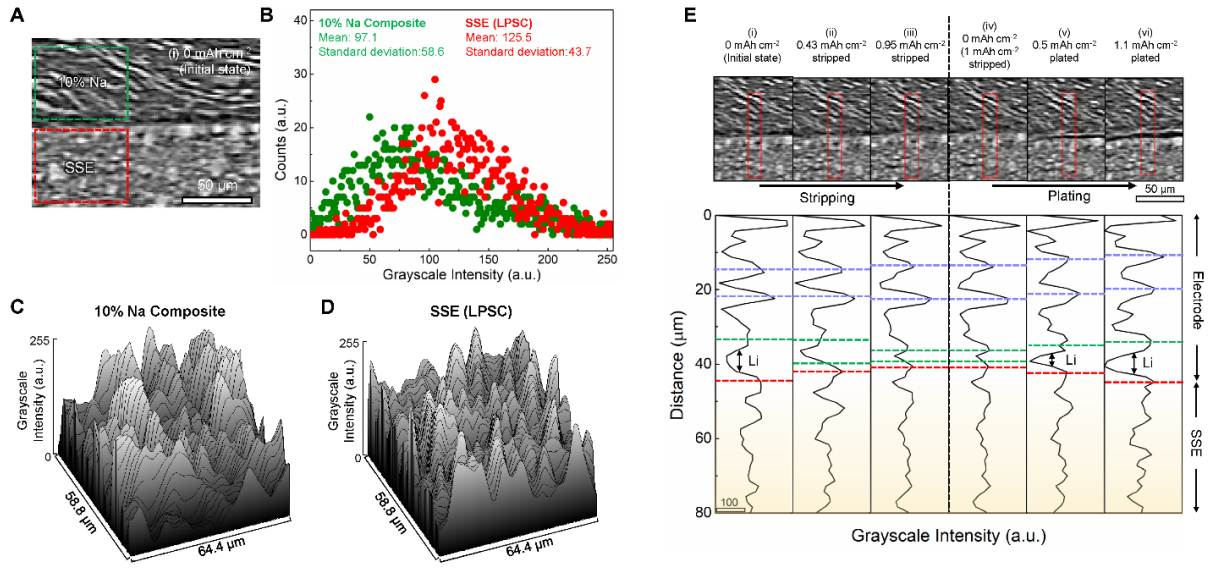


Fig. S15. Operando X-ray computed tomography analysis. (**A**) Cross-sectional reconstructed XCT image slice of the Li/Na electrode-SSE interface, captured at a different location than the analysis in the main text in Fig. 3A, B. (**B-D**) Grayscale intensity analysis of the electrode and SSE. (B) Histogram of grayscale intensity for the Li/Na composite electrode (green) and SSE (red) regions shown in (A). (C, D) 2D planar distributions of grayscale intensity of the composite electrode (green box) and the SSE (red box) regions. The average grayscale intensities were 97.1 (±58.6) for the composite electrode and 125.5 (±43.7) for the SSE. (**E**) Cross-sectional image slices at different stages of electrochemical cycling (top) and corresponding line-scanned grayscale profiles (bottom) in the red-boxed region. This analysis is for a different location than that in the main text in Fig. 3. The stages of electrochemical cycling, from left to right, are: (i) 0 mAh cm^{-2} stripped (initial state), (ii) 0.43 mAh cm^{-2} stripped, (iii) 0.95 mAh cm^{-2} stripped, (iv) 0 mAh cm^{-2} plated (equivalent to 1 mAh cm^{-2} stripped), (v) 0.5 mAh cm^{-2} plated, and (vi) 1.1 mAh cm^{-2} plated. In the grayscale profiles, SSE interfaces are indicated by red lines, while the nearest Na phase to the SSE interface and the two Na peaks subsequent to the first one are marked by green and blue lines, respectively.

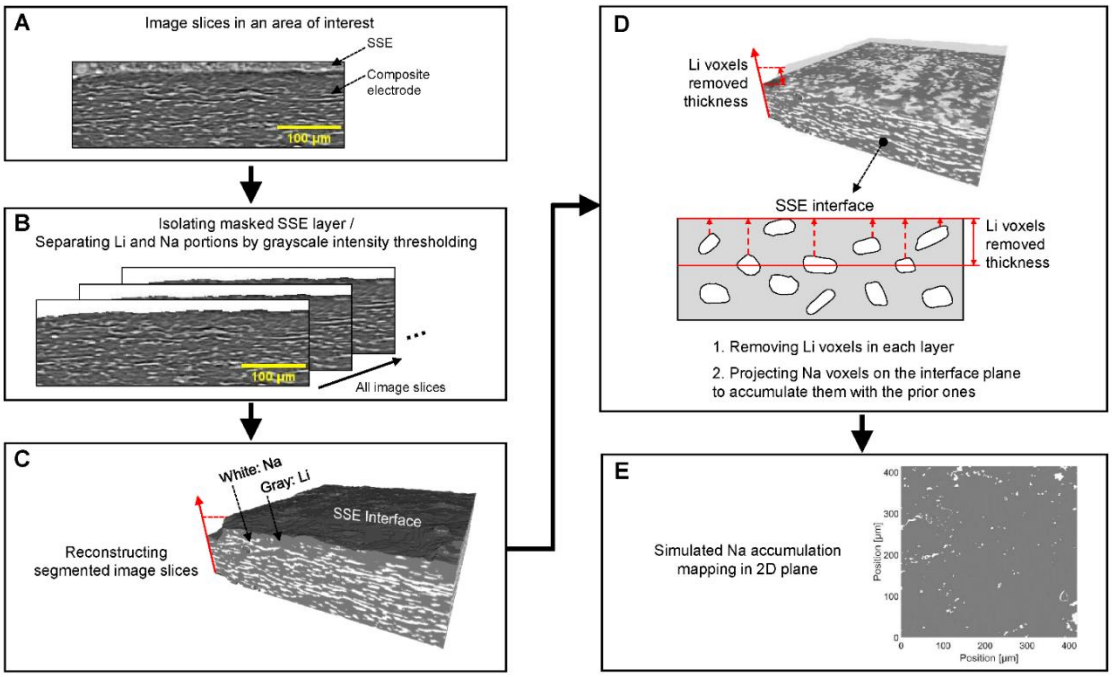


Fig. S16. A schematic of the XCT data segmentation procedure used for the "simulated Li stripping" analysis presented in Fig. 3C-E. (**A**) The process involves segmenting image slices in the region of interest along the entire interface. (**B**) To reconstruct the segmented Li/Na composite volume, the SSE layer was masked, and the composite electrode layer was isolated using grayscale intensity differences, shown in Fig. S15. (**C**) Li and Na portions in the image slices were distinguished by applying a threshold to the grayscale intensity, constructing a subvolume segmented into binary components corresponding to Li and Na phases. (**D**) Simulated stripping was then conducted by selectively removing Li voxels from the plane adjacent to the SSE interface. (**E**) The remaining Na voxels in this plane were subsequently projected onto a virtual SSE interface. This process was then repeated for nine additional voxel planes, allowing for simulated mapping of Na accumulation at the virtual SSE interface, as shown in Fig. 3E. The interfacial coverage changes of Li and Na, shown in Fig. 3D, were calculated using this 2D mapping. The simulated mapping of the first and second virtual stripping steps in Fig. 3D was conducted using the reconstructed subvolumes at (i) 0 mAh cm^{-2} (the initial state before any stripping) and (vi) 1.1 mAh cm^{-2} plating (after stripping and plating in the experiment), as shown in Fig. 3B.

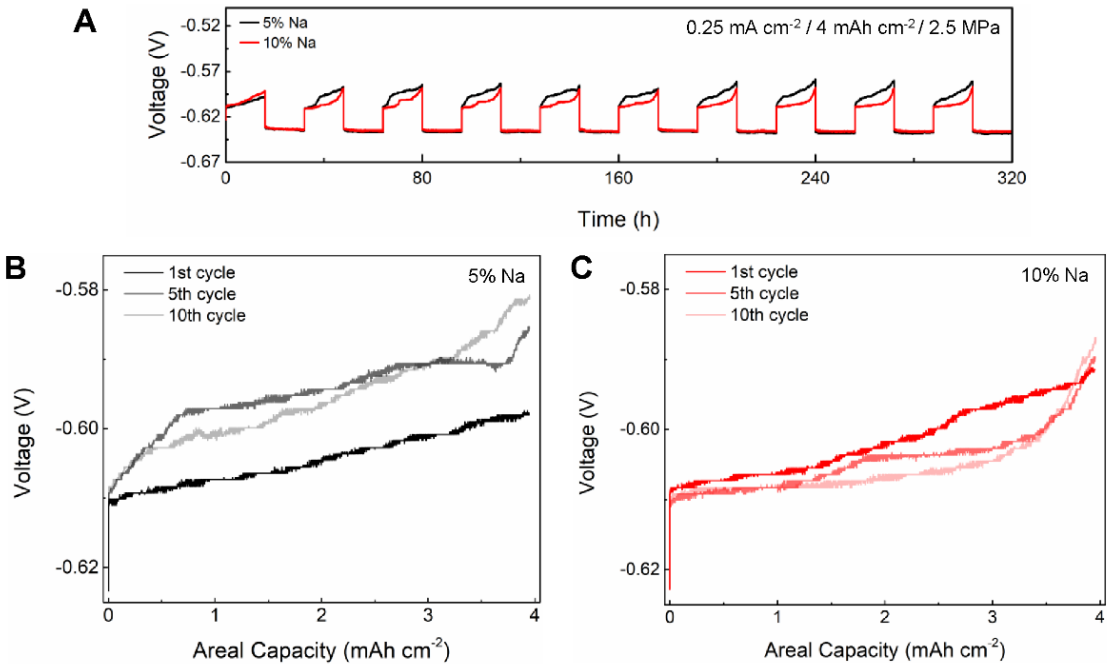


Fig. S17. Galvanostatic cycling of the 5% and 10% Na electrodes. (**A**) Li cycling voltage profiles with 4 mAh cm^{-2} capacity per cycle at a current density of 0.25 mA cm^{-2} and a stack pressure of 2.5 MPa. Li-In was the counter electrode in the tests. (**B**, **C**) Li stripping voltage profiles from the 1st, 5th, and 10th cycles of the (B) 5% Na and (C) 10% Na electrodes. In the first cycle, the 10% Na electrode showed more rapid voltage polarization. However, after the first cycle, the stripping voltage profiles of the 5% Na electrode polarized faster compared to the 10% Na electrode, indicating that the higher Na content in the 10% Na electrode provided a better capability for stabilizing the Li-SSE interface.

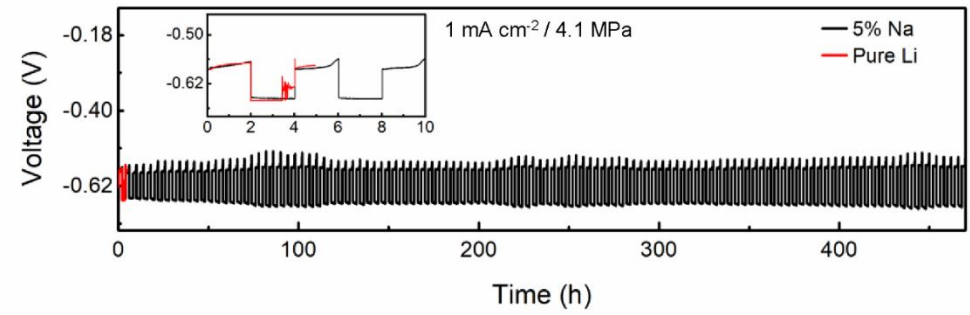


Fig. S18. Galvanostatic cycling of 5% Na and pure Li electrodes at a current density of 1 mA cm^{-2} and a stack pressure of 4.1 MPa. Li-In electrodes were used as counter electrodes, and the cells were cycled with a capacity of 2 mAh cm^{-2}. The pure Li electrode was short circuited in the first cycle.

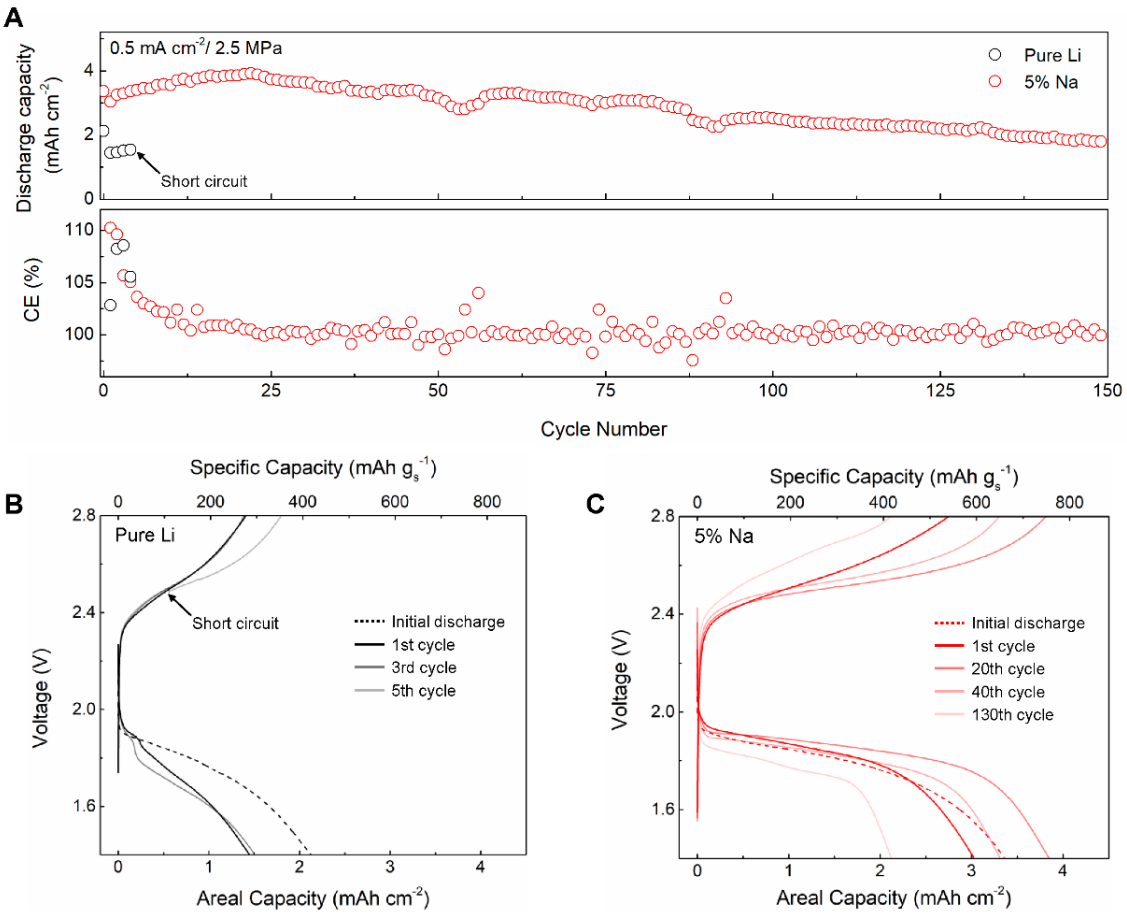


Fig. S19. Galvanostatic cycling tests of 5% Na and pure Li anodes in full cells with sulfur composite cathodes. (**A**) Li cycling behavior of 5% Na and pure Li anodes with sulfur composite cathodes at a current density of 0.5 mA cm^{-2} (~0.06 C, 1C=1657 mA g_s^{-1}) and a stack pressure of 2.5 MPa. (**B**, **C**) voltage profiles of (B) the pure Li anode and (C) the 5% Na anode. The cathode composite loading was 15.3 mg cm^{-2}, corresponding to ~8.5 mAh cm^{-2} theoretical capacity. 6 mg cm^{-2} of anode material was loaded in both cells. The cells were cycled over a voltage range of 1.4 to 2.8 V. The discharge capacities in the first cycle in (A) show the initial discharge capacities of the pure Li and 5% Na cells. The dashed lines in (B) and (C) denote the initial discharge profiles of each cell.

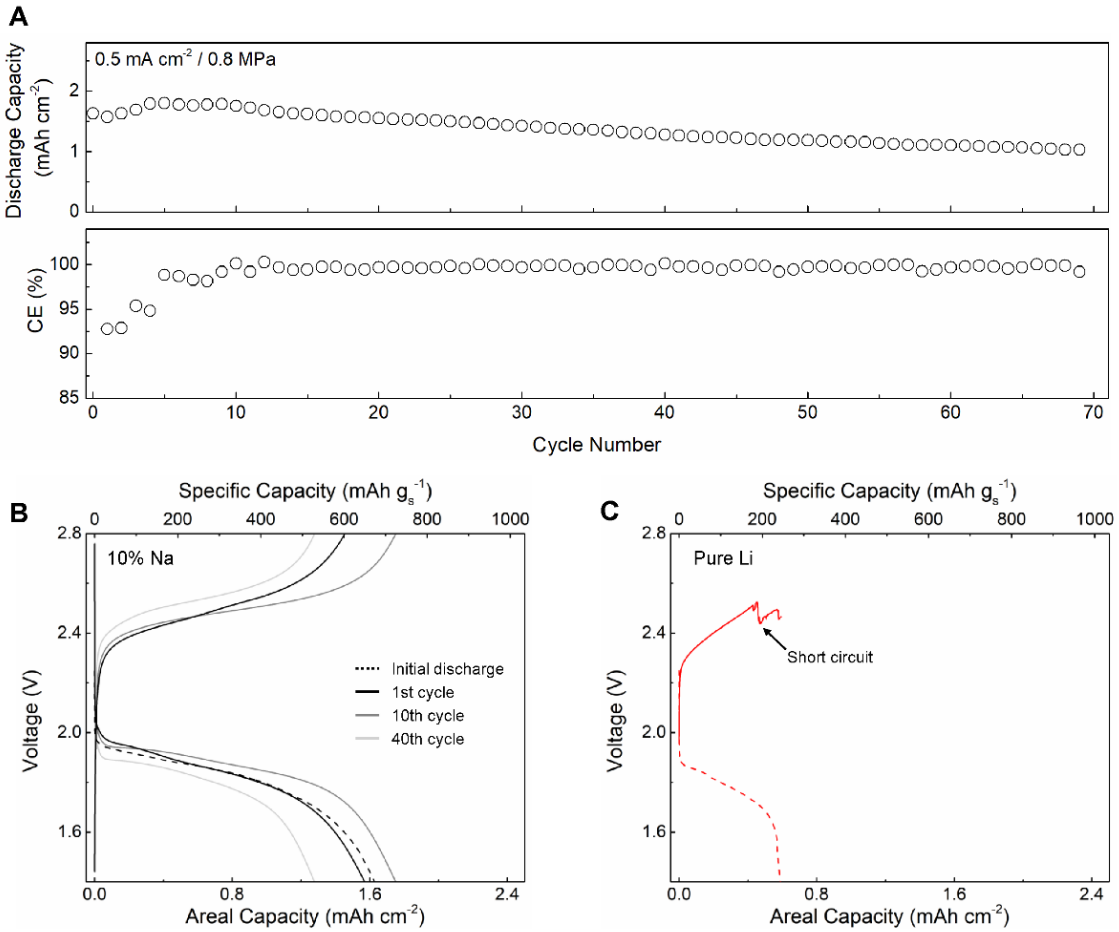


Fig. S20. Galvanostatic cycling tests of 10% Na and pure Li anodes in full cells with sulfur composite cathodes. (**A**) Cycling behavior of a 10% Na anode in a full cell assembled with a sulfur composite cathode. The cell was cycled at a current density of 0.5 mA cm^{-2} and a stack pressure of 0.8 MPa. (**B**, **C**) Voltage profiles of cells with (B) 10% Na, and (C) pure Li anodes. The dashed lines in (B) and (C) denote initial discharge profiles. The pure Li anode was short circuited in the first cycle after the initial discharge. ~7.3 mg cm^{-2} of composite cathode was loaded in both cells, corresponding to ~4.1 mAh cm^{-2} theoretical capacity. 5.7 mg cm^{-2} of anode material was loaded in both cells. The cells were cycled over a voltage range of 1.4 to 2.8 V.

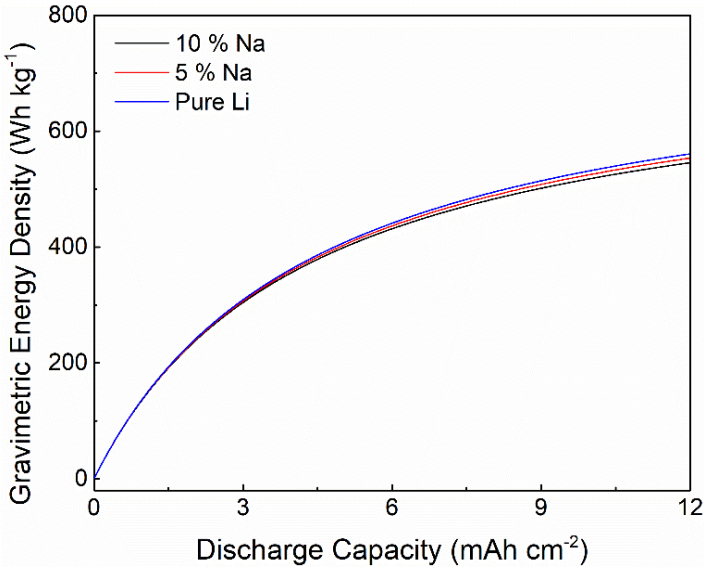


Fig. S21. Gravimetric energy density (Specific energy) as a function of areal discharge capacity using three different anodes. The calculation was conducted for solid-state sulfur batteries with composite anodes containing 0% Na (pure Li), 5% Na, and 10% Na. Incorporating Na metal into the Li anode had a negligible effect on the gravimetric energy density of the cells. Calculation details are presented in Table S5.

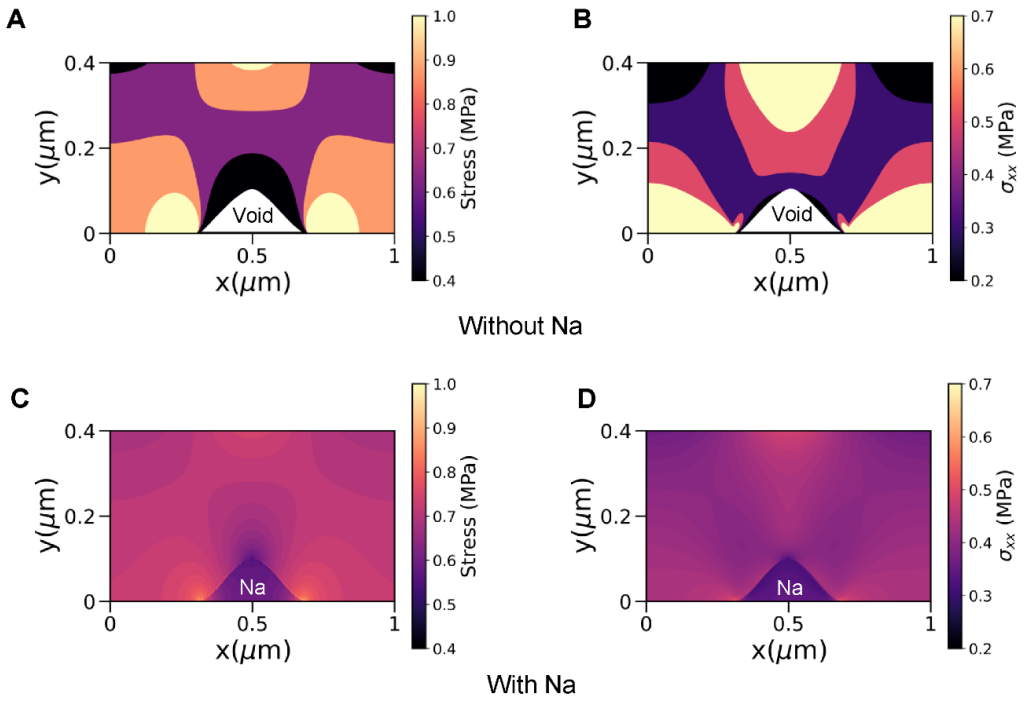


Fig. S22. Stress distributions in a void and a Na domain within Li anode. (**A**) Hydrostatic stress and (**B**) normal stress along the *x*-direction in Li metal with a void at the Li-SSE interface (void size, w/L = 0.3). (**C**) Hydrostatic stress and (**B**) normal stress along the *x*-direction in Li metal with a Na domain at the Li-SSE interface (Na domain size, w/L = 0.3).

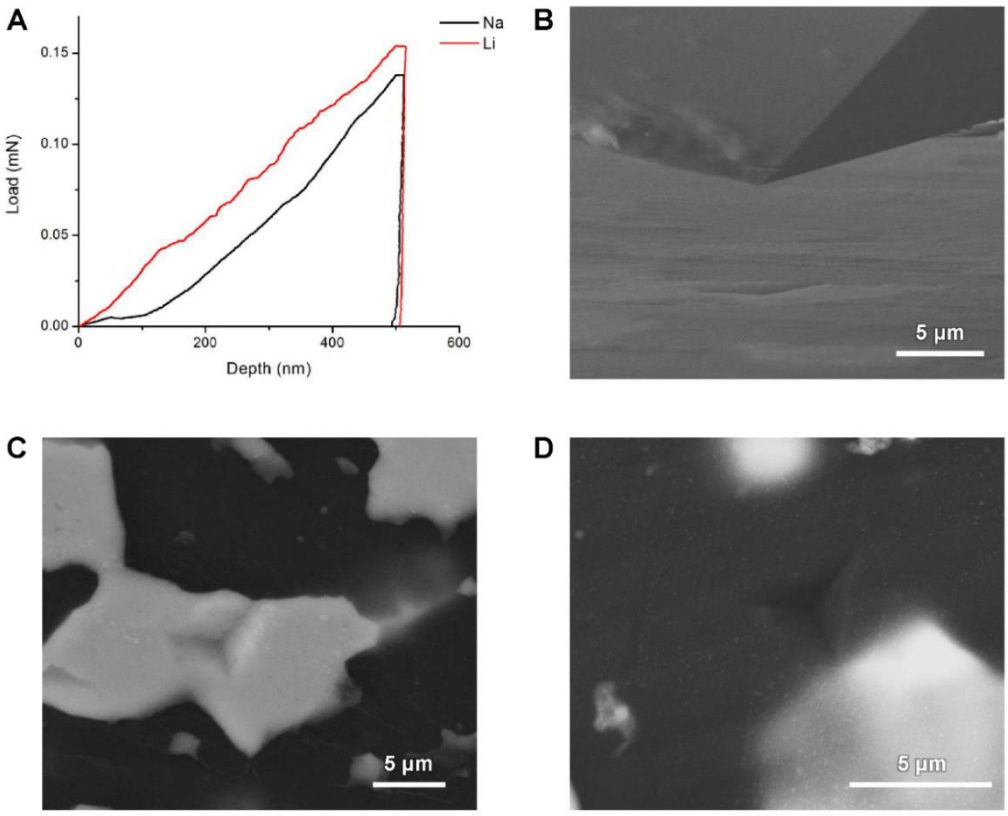


Fig. S23. Nanoindentation tests of 20% Na foil. (**A**) Load-depth profiles from nanoindentation tests on Li and Na regions on a 20% Na foil. (**B**) SEM image during the nanoindentation experiment on the Na phase. 85˚-tilted view of the Berkovich tip hovering 6 µm above the sample surface with residual nanoindentation impression on Na region. Top-view SEM images of residual nanoindentation impressions on (**C**) a Na region and (**D**) a Li region.

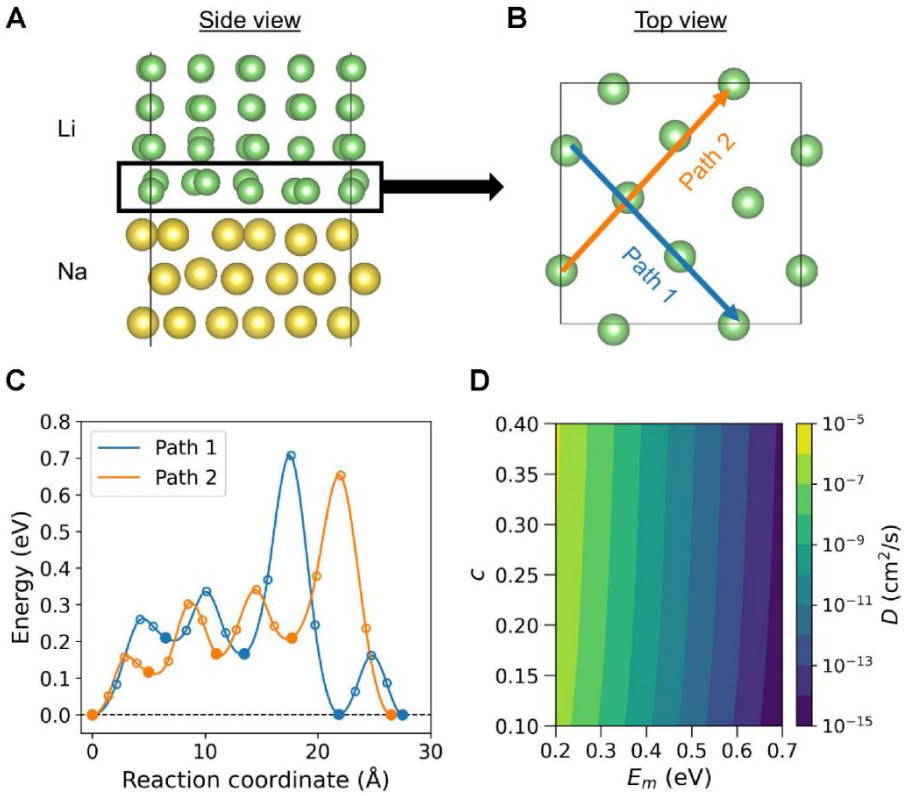


Fig. S24. DFT modeling of diffusion. (**A**) Side view of the Li(100)-Na(100) interface model. (**B**) Top view of the Li layer at the interface, showing the two migration pathways of Li vacancies considered. (**C**) Energy landscape of the two migration pathways considered. Closed circles correspond to configurations where the vacancy is located at a Li site, and open circles correspond to intermediate NEB images. (**D**) 300 K interface diffusivity as a function of migration barrier energy and vacancy site fraction.

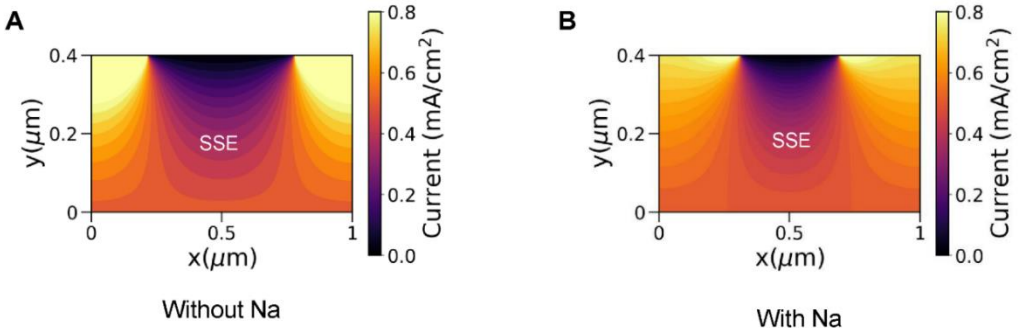


Fig. S25. Current distribution in the SSE at time t = 2 min (**A**) without and (**B**) with Na at the Li-SSE interface.

Tables

Table S1. Calculation details for determining the stack plates mass and cell mass for Fig. S1.

Stack plates mass calculation		
Parameter	Value	Notes
Stack plates mass [g]	$= 2 \cdot A \cdot d \cdot (\frac{aPL^4}{yE})^{1/3}$	P is stack pressure in this equation (*88*)
Stack plate size (A) [cm^2]	12	3 (L) × 4 (W) cm^2, where L and W are width and length, respectively
Plate density (d) [g cm^{-3}]	7.85	Material: Steel
Plate constant (a)	0.15	Poisson ratio: 0.3
Plate length (L) [m]	0.03	The shorter dimension of the plate
Deflection tolerance (y) [m]	6×10^{-5}	Assuming y= L/5000, L/500, and L/100
Elastic modulus of plate (E) [N m^{-2}]	2×10^{11}	

Cell mass calculation		
Parameter	Value	Notes
Active cathode material	$LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC 811)	
Composite cathode mixing ratio	NMC811: 70wt.% SSE: 27.5 wt.% Carbon: 2.5 wt.%	(*89*)
Pressed density [g cm^{-3}]	3.6	
Areal capacity [mAh cm^{-2}]	4	
Cathode areal density [g cm^{-2}]	2.9×10^{-2}	
SSE material	Li_6PS_5Cl	
Press density [g cm^{-3}]	1.87	Assuming zero porosity
Thickness [μm]	30	
SSE areal density [g cm^{-2}]	5.6×10^{-3}	
Cathode current collector	Al 16 μm	
Areal density [g cm^{-2}]	4.3×10^{-3}	Al density: 2.7 g cm^{-3}
Anode	Li 20 μm / Cu 8 μm	
Anode areal density [g cm^{-2}]	8.2×10^{-3}	Cu density: 8.96 g cm^{-3} Li density: 0.53 g cm^{-3}

Table S2. A list of references and other information for the data presented in Fig. S2.

Active material	Stack pressure [MPa]	Current density [mA cm^{-2}]	Cycling capacity [mAh cm^{-2}]	Cell structure	Reference
Li metal	3	0.2	1	Symmetric cell	(*5*)
	7	1	1	Symmetric cell	
	15	2.2	2	Symmetric cell	(*90*)
	5	0.1	1	Li‖NCA	(*91*)
	15	0.5	1.5	Cu‖Li (pre-deposited excess Li)	(*92*)
LiAg	5	0.1	1	Symmetric cell	(*93*)
Ag, Au	15	0.25	1	Ag, Au‖Li	(*25*)
Li-In	150	3.8	4	Li-In‖NMC	(*86*)
Al/In	50	6.5	2.5	Al/In‖NMC	(*89*)
In	8	1	5	In‖NMC	(*28*)
Al, Sn, Si	8	0.5	5	Al, Sn, Si‖NMC	
Si	50	5	5	Si‖NMC	(*94*)
Si	20	0.24	2.4	Si‖NMC	(*95*)
LiSi	75	5	4	LiSi‖LCO	(*96*)
Mg	20	0.65	6.5	Mg‖NMC	(*97*)
Mg	30	0.5	2	Mg‖NMC	(*98*)
LiMg	5	0.31	N/A	Symmetric cell	(*24*)
Na 5 at.%	2.5	0.75	2	Li/Na‖Li-In	This study
Na 5 at.%	4.1	1	2	Li/Na‖Li-In	
Na 10 at.%	1.6	0.5	2	Li/Na‖Li-In	
Na 5 at.%	2.5	0.5	~4	Li/Na‖S	
Na 5, 10 at.%	0.8	0.5	~1.5	Li/Na‖S	

Table S3. Material properties of Li and Na metal, as well as LPSC.

	Elastic modulus [GPa]	Yield strength [MPa]	Homologous temperature (25 °C)	Redox potential [V] (*vs.* SHE)	Reference
Li metal	7.8	0.73-0.81[a]	0.66	-3.04	(*46*, *47*)
Na metal	4.6	0.16–0.24[b]	0.80	-2.71	(*46*, *47*)
LPSC	22		< 0.3[c]		(*52*, *99*)

a. Li metal – bulk tension @ $\dot{\varepsilon}$ (strain rate) ~ 10^{-3}

b. Na metal – bulk compression @ $\dot{\varepsilon}$ (strain rate) ~ 10^{-3}

c. Li_6PS_5Cl is thermally stable over 700 °C. Its homologous temperature is less than 0.3, suggesting that there is no creep behavior at room temperature.

Table S4. Parameters of fitted lines shown in fig. S6B and interfacial transport descriptors estimated from Sand's equation in Supplementary Text 1.

Electrode	Slope	R^2	Interfacial transport descriptor [$\times 10^{-11}$ cm^2 S^{-1}]
Pure Li	2.80	0.98	0.67
2.5% Na	2.56	0.98	0.84
5% Na	2.44	0.99	0.98
10% Na	1.96	0.97	1.68
20% Na	1.46	0.98	2.47

Table S5. Calculation of specific energies for cells containing Li and Li/Na anodes and sulfur-based cathodes. 5 mAh cm^{-2} capacity was assumed in this calculation. Figure S21 shows the trend of specific energy as a function of areal capacity in the cell.

Cell component	Parameter	Value	Note
Cathode	Total cathode mass (m_c) [mg cm^{-2}]	11.1	
	Mass ratio = SSE : Carbon : Sulfur	3:1:2	
	Sulfur loading [mg cm^{-2}]	3.70	
	Specific capacity [mAh g_s^{-1}]	1350	Reported capacity in 1-2.7 V range (*43*)
	Areal capacity (Q_d) [mAh cm^{-2}]	5	
	Average discharge voltage (V_{ad})[V]	1.91	(*100*)
SSE	SSE mass (m_{sse}) [mg cm^{-2}]	6.80	
	LPSC density [g cm^{-3}]	1.89	
	Porosity [%]	10	
	Thickness [µm]	40	
Current collector	Total current collector mass (m_{cc}) [mg cm^{-2}]	4.29	Ni 6 µm / Al 12 µm
Anode (Case 1)	**Na concentration [at.%]**	**10**	
	Total anode mass (m_{a-1}) [mg cm^{-2}]	1.77	Mass ratio: 73.1% Li and 26.9% Na
	Li metal mass [mg cm^{-2}]	1.29	
	Na metal mass [mg cm^{-2}]	0.48	
	Anode areal capacity [mAh cm^{-2}]	5	Anode/cathode capacity ratio = 1:1
	Total cell mass [mg cm^{-2}]	23.97	$= m_c+m_{sse}+m_{cc}+m_{a-1}$
	Specific energy [Wh kg^{-1}]	**398.35**	$= Q_d \times V_{ad} / (m_c+m_{sse}+m_{cc}+m_{a-1})$
Anode (Case 2)	**Na concentration [at.%]**	**5**	
	Total anode mass (m_{a-2}) [mg cm^{-2}]	1.52	Mass ratio: 85.2% Li and 14.8% Na
	Li metal mass [mg cm^{-2}]	1.29	
	Na metal mass [mg cm^{-2}]	0.23	
	Anode areal capacity [mAh cm^{-2}]	5	Anode/cathode capacity ratio = 1:1
	Total cell mass [mg cm^{-2}]	23.72	$= m_c+m_{sse}+m_{cc}+m_{a-2}$
	Specific energy [Wh kg^{-1}]	**402.55**	$= Q_d \times V_{ad} / (m_c+m_{sse}+m_{cc}+m_{a-2})$
Anode (Case 3)	**Na concentration [at.%]**	**0**	
	Total anode mass (m_{a-3}) [mg cm^{-2}]	1.29	
	Li metal mass [mg cm^{-2}]	1.29	
	Na metal mass [mg cm^{-2}]	0	
	Anode areal capacity [mAh cm^{-2}]	5	Anode/cathode capacity ratio = 1:1
	Total cell mass [mg cm^{-2}]	23.50	$= m_c+m_{sse}+m_{cc}+m_{a-3}$
	Specific energy [Wh kg^{-1}]	**406.42**	$= Q_d \times V_{ad} / (m_c+m_{sse}+m_{cc}+m_{a-3})$

Table S6. Experimental parameters (A_c and P) and measured hardness (H) of Li and Na phases shown in fig. S23.

	Contact area (A_c) [μm^2]	Indentation load (P) [mN]	Hardness (H) [MPa]
Li	9.89	0.158	16
	9.08	0.154	17
Na	16.77	0.138	8.2
	19.32	0.134	9.6

Table S7. Calculated Li diffusivities and migration barriers on Li surface and Li-Na interfaces.

	Calculated diffusivity [$cm^2\ s^{-1}$]	Migration barrier [eV]	Reference
Li surface (Li(100))	$\sim10^{-3}$ -$\sim10^{-12}$	0.02-0.47	(*75*, *78*)
Li-Na interface	$\sim10^{-6}$ -$\sim10^{-15}$	0.2-0.7	This study

Table S8. List of parameters and properties used in the electro-chemo-mechanical model.

	Parameters	Values	Units
L	Domain width	1	μm
k_e	Ionic conductivity of electrolyte	0.01	mS cm^{-1}
E_{Na}	Young's modulus of Na	4	GPa
E_{Li}	Young's modulus of Li	7.8	GPa
$\sigma_{0_{Na}}$	Yield strength of Na	0.3	MPa
$\sigma_{0_{Li}}$	Yield strength of Li	0.8	MPa
Ω_{Li}	Partial molar volume of Li	13	$cm^3\ mol^{-1}$
i_0	Exchange current density	1	mA cm^{-2}
H	Hardening Modulus	50	MPa
p_{ext}	Stack pressure	1	MPa
A_{Na}	Effective creep constant for Na	0.022	s^{-1}
A_{Li}	Effective creep constant for Li	0.00029	s^{-1}
n_{Na}	Stress exponent for Na	5	-
n_{Li}	Stress exponent for Li	6.6	-
D_{Li}	Li diffusivity on Li surface	10^{-7}	$cm^2\ s^{-1}$
$D_{Li,Na}$	Li diffusivity on Li/Na interface	10^{-9}	$cm^2\ s^{-1}$
δ_b	Interface thickness	1	nm
F	Faraday constant	96,485	C mol^{-1}
T	Temperature	298	K
R	Universal Gas Constant	8.314	J $mol^{-1}\ K^{-1}$

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